

ANIMAL HEALTH

Level – IV

**Based on March, 2022 Version -4 Occupational
Standard**



**Module Title: Applying Veterinary Drug and
Chemicals**

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Table of Contents

Introduction to the Module	1
LO #1-Acquire basic knowledge in pharmacology	2
Instruction sheet 1	2
Information Sheet 1	4
Self-check 1	64
LO #2- Apply therapeutics drugs based on causative agent and body system	67
Instruction sheet 2	67
Information Sheet 2	69
Self-check 2	115
LO #3-Identify and administer biological fluids	117
Instruction sheet 3	117
Information Sheet 3	118
Self-check 3	122
LO #4-Store and dispense veterinary drugs and chemicals	124
Instruction sheet 4	124
Information Sheet 4	125
Self-check 4	127
LO #5-Apply stock control and clinic security	129
Instruction Sheet 5	129
Information Sheet 5	130
Self-check 5	133
Reference Materials	135

Introduction to the Module

This module covers the knowledge, skill and attitude required to use, control, prescribe, administer veterinary drug and manage associated risk management procedures according to organizational guidelines and procedures in veterinary service. It also covers the knowledge and skill required to store, dispense veterinary drug/chemical as well as apply activities of stock control and clinic security.

LG #1	LO #1-Acquire basic knowledge in pharmacology
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Instruction sheet 1
This learning guide is developed to provide you the necessary information regarding the following content coverage and topics: • Introduction to pharmacology and rational drug use • Causes, characteristics and implications of irrational drug use • Basic information on nomenclature of drugs • Veterinary drug dosage forms, dose calculations and routes of administration • Drug prescription and pharmacological abbreviations • The fate and mechanisms of actions of drugs • Pharmacokinetics and pharmacodynamics of veterinary drugs This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to: • Mention and describe technical terms in pharmacology and rational drug use. • Mention characteristics of irrational drug use. • Mention and describe causes and implications associated with irrational drug use. • Gain general information on nomenclature of drugs. • Mention dosage forms of drug and explain dose calculations of veterinary drug. • Mention routes of drug administration in different species of animals for different types of drugs. • Describe drug prescription and pharmacological abbreviations in prescription. • Explain the concept of pharmacokinetics and dynamics of veterinary drugs.
Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

Information Sheet 1

1.1.Introduction to Pharmacology and Rational Drug Use

1.1.1. Definition of terms related with pharmacology

- **Pharmacology:** The term pharmacology is derived from two Greek words pharmakon meaning “drugs, medicine or poison”; and logos, “study or science” Thus, pharmacology is defined as “the Study of Drugs or the Science of Drugs”
- **Drug:** A drug can be defined as a chemical substance of known structure, other than a nutrient or an essential dietary ingredient, which when administered to a living organism, produces a biological effect.
- **Medicine:** A chemical preparation which usually but not necessarily contains one or more drugs, administered with the intention of producing a therapeutic effect , and usually contain other substances (excipients, stabilizers, solvents etc.) besides the active drug to make them more convenient to use.
- **Antimicrobials** are agents which are used against microorganism, either to inhibit their growth or multiplication, or to kill them.
- **Antibiotics** are the agents which are usually produced by non-pathogenic microorganism (bacteria, fungi) and are used for either killing or inhibiting the growth of other pathogenic microorganism, without affecting the host tissue.
- **Pharmacokinetics** deals with the absorption, distribution, and excretion of drugs.
- **Pharmacodynamics:** It refers to the study of response of an organism to the action of drugs in absence of disease.
- **Chemotherapy:** It is a branch of pharmacology dealing with drugs that selectively inhibit or destroy specific agents or disease such as bacteria, viruses, fungi and other parasites.
- **Definitive therapy:** Permitted only for the definitive diagnosis as proven infection responsible pathogenic microorganism is already known.
- **Empirical therapy:** drugs are used based on past experience with their actions rather than scientific evidence

- **Prophylactic or preventive therapy:** It is the therapy aimed at preventing the occurrence of a disease, like the use of vaccines for preventing bacterial and viral infections.
- **Symptomatic or palliative therapy:** It aims at treating the condition based on the symptoms and providing relief to the patient without actually spending time on finding the cause of the disease.
- **Supportive therapy:** drug therapy that help to maintain body functions -- fluid and electrolytes, volume expanders.
- **Toxicology:** It is defined as the study of poisons. It includes study of toxicity or adverse effects of drugs or chemical, physical or biological agents in man and animals.
- **Dose** is the amount of drug taken at any one time. It can be expressed as the weight of drug
- **Dosage regimen** is the frequency at which the drug doses are given..
- **Optimal dosage** is the dosage that gives the desired effect with minimum side effects.

1.1.2. Rational veterinary drug use

- It is defined as that sick animal should receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them.

It includes:-

- Steps in defining patients' problems (or diagnosis);
- Defining effective and safe treatments (drugs and non-drugs);
- Selecting appropriate drugs, dosage and duration;
- Writing a prescription; in giving patients adequate information;
- Planning to evaluate treatment responses

1.2.Causes, characteristics and implications of irrational drug use

- Irrational drug use: refers to the misuse, abuse, or inappropriate use of medications or drugs.

1.2.1. Causes of irrational use of medicines:

- Lack of information, training and education,
- Poor communication between veterinarian & owner,
- Lack of diagnostic facilities,
- Promotional activities of pharmaceutical industries,
- The use of too many drugs per patient (poly pharmacy),
- Inappropriate use of antimicrobials, often in inadequate dosage,
- Over-use of injections when oral formulations would be more appropriate,
- Failure to prescribe in accordance with clinical guidelines,
- Inappropriate self-medication, often of prescription only drugs.

1.2.2. Characteristics of irrational drug use:

- Over prescription,
- Omission,
- The use of inappropriate dosage,
- Incorrect duration,
- Misuse of drug,
- Frequent use of drugs and
- Not stick with withdrawal period of a drug

1.2.3. Implications of irrational drug use:

- Maximize the cost of drug therapy,
- Inability to avoid preventable drug interactions,
- Risks of adverse drug reactions (ADRs) is increased,
- Reduce the quality of therapeutic care while promoting patient adherence,
- Low quality of drug therapy that could increase patients' morbidity and mortality,
- Leads to the emergence of drug resistance.

1.2.4. Interventions to reduce irrational drug use:

- Use clinical guidelines
- Supervision, audit and feedback
- Independent information on drugs
- Public education about drugs
- Appropriate and enforced regulation
- Sufficient government expenditure to ensure availability of drugs and staff.

- Elimination of financial incentives that lead to improper prescribing, such as prescribers selling medicines for profit to supplement their income.

1.3. Basic information on nomenclature of drugs

There are three types of drug names:

A. Chemical name:

- Indicate the drug's chemical structure
- Is a scientific name that precisely describes its atomic and molecular structure.
- It is long, complex and inappropriate for everyday use.
- IUPAC name

B. Generic (Official or Approved or Trivial name):

- Preferred use: shortened form of chemical name
- The chemical compound is known throughout the world by this generic name.
- Approved names are generally used by researchers and non-clinical teachers.
- Is an abbreviation of the chemical name but it is less complex.
- Each drug has only one generic name.

C. Proprietary (Brand or Trade name):

- One chemical compound can have several proprietary names.
- Even one proprietary name may not contain single chemical.
- Manufacturers and clinicians prefer to use brand names.
- Created by the drug company that manufactures the drugs.

Table 1.1: Summary of drug nomenclature

Chemical name	Generic name	Trade name
3,4-dihydroxyphenylethanolmethamphetamine	Epinephrine	Adrenaline
Acetylsalicylic acid	Aspirin	Ecosprin
Propionic acid	ibuprofen	Motrin ® , Profen ®

Note: Individual drugs may have several different names, but the two that are most commonly used are the generic name and the trade or brand name. To avoid confusion, it is best to use a drug's generic name because any one drug can have a number of trade names.

1.4. Veterinary drug dosage forms, dose calculations and routes of administration

1.4.1. Veterinary drug dosage forms

Veterinary drug dosage forms can be:-

A. Oral drug dosage forms

Oral dosage forms comprise liquids (solutions, suspensions, and emulsions), semi-solids (pastes), and solids (tablets, capsules, powders, granules, premixes, and medicated blocks).

- **Solution**

- ✓ A mixture of 2 or more components that form a single phase that is homogeneous down to the molecular level.
- ✓ Solutions are absorbed faster and generally cause less irritation of the GI mucosa.
- ✓ Susceptibility to microbial contamination,
- ✓ Hydrolysis in aqueous solution of susceptible active ingredients, and
- ✓ Taste of some drugs is more unpleasant when in solution

- **Suspension**

- ✓ A coarse dispersion of insoluble drug particles, generally with a diameter exceeding 1 μm , in a liquid (usually aqueous) medium
- ✓ It is useful for administering insoluble or poorly soluble drugs or in situations when the presence of a finely divided form of the material in the GI tract is required.

- **Emulsion**

- ✓ A system consisting of 2 immiscible liquid phases, one of which is dispersed throughout the other in the form of fine droplets; droplet diameter generally ranges from 0.1-100 μm .

- **Paste**

- ✓ A 2-component semi-solid in which drug is dispersed as a powder in an aqueous or fatty base.
- ✓ The particle size of the active ingredient in pastes can be as large as 100 μm .
- ✓ The vehicle containing the drug may be water.

- ✓ Can be easily and safely administered by owners.
- **Tablet**
 - ✓ Consists of one or more active ingredients and numerous excipients
- **Capsule**
 - ✓ An oral dosage form usually made from gelatin and filled with an active ingredient and excipients.
- **Powder**
 - ✓ A formulation in which a drug powder is mixed with other powdered excipients to produce a final product for oral administration.
- **Granule**
 - ✓ Dosage form consisting of powder particles that have been aggregated to form a larger mass, usually 2-4 mm in diameter.

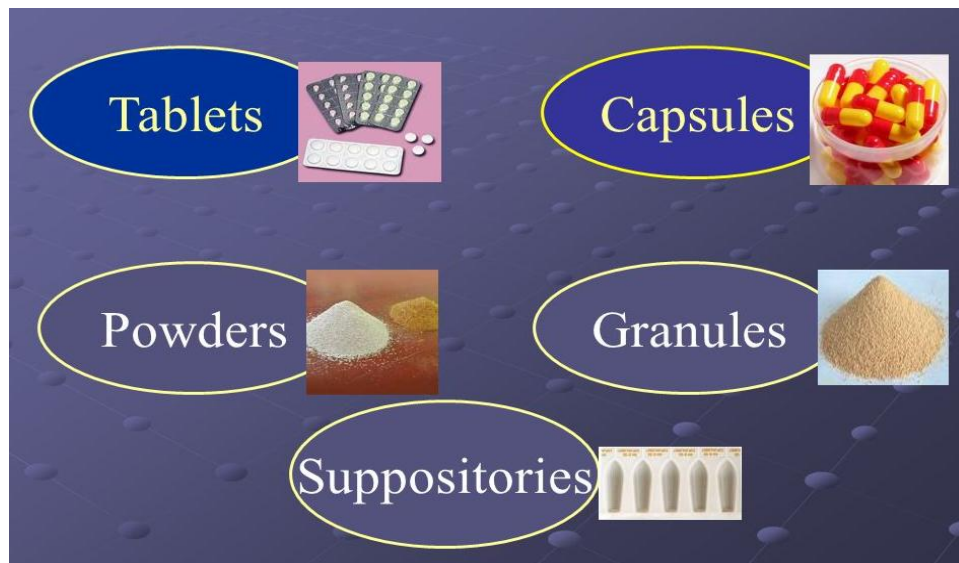


Figure 1.1: Solid dosage forms

B. Parenteral dosage forms

Delivery systems include injectable (ie, solutions, suspensions, emulsions, and dry powders for reconstitution), intra-mammary infusions, intra-vaginal delivery systems, and implant

- **Solution**

- ✓ For injection is a mixture of 2 or more components that form a single phase that is homogeneous down to the molecular level
- ✓ “Water for injection” is the most widely used solvent for parenteral formulations.

- **Suspension**

- ✓ For injection consists of insoluble solid particles dispersed in a liquid medium, with the solid particles accounting for 0.5-30% of the suspension.
- ✓ The vehicle may be aqueous, oil, or both. Injectable suspensions are commonly used.

- **Emulsion**

- ✓ For injection is a heterogeneous dispersion of one immiscible liquid in another; it relies on an emulsifying agent for stability.
- ✓ Parenteral emulsions are rare because it is seldom necessary to achieve an emulsion for drug administration.

- **Dry powder**

- ✓ For parenteral administration is reconstituted as a solution or as a suspension immediately prior to injection.
- ✓ The principal advantage of this dosage form is that it overcomes the problem of instability in solution.

- **Mastitis intra-mammary infusion products**

- ✓ Available for lactating and non-lactating (dry) cows.
- ✓ Lactating cow intra-mammary infusions should demonstrate fast and even distribution of the drug and a low degree of binding to udder tissue.
- ✓ These properties result in lower concentrations of drug residues in the milk.

C. Topical dosage forms

It is available for treating animals include solids (dusting powders), semisolids (creams, ointments, and pastes), and liquids (solutions, suspension concentrates, suspoemulsions, and emulsifiable concentrates).

- **Dusting powder**

- ✓ A finely divided insoluble powder containing ingredients such as talc, zinc oxide, or starch
- **Cream**
 - ✓ A semisolid emulsion formulated for application to the skin or mucous membranes. Droplet diameter in topical emulsions generally ranges from 0.1-100 μm .
- **Ointment**
 - ✓ A greasy, semisolid preparation that contains dissolved or dispersed drug.
 - ✓ Indicated for chronic, dry lesions and contraindicated in exudative lesions.
- **Paste**
 - ✓ For topical use a stiff preparation containing a high proportion of finely powdered solids such as starch, zinc oxide, calcium carbonate, and talc.
 - ✓ Less greasy than ointments because much of the fluid hydrocarbon fraction is absorbed onto the solid particles.
 - ✓ Indicated for ulcerated lesions.
- **Solution**
 - ✓ For topical use is a mixture of 2 or more components that form a single phase down to the molecular level.
 - ✓ Topical solutions include eye drops, ear drops, and lotions.
- **Suspension**
 - ✓ Concentrate for topical use is a mixture of insoluble, solid active ingredients, which are normally at high concentrations, in water or oil.
- **Suspension**
 - ✓ Combines the elements of an emulsion and a suspension, allowing active ingredients with widely varying physical properties to be formulated in a single product.
- **Liniments**
 - ✓ Applied to the skin to alleviate pain, provide relief from muscle soreness, and reduce inflammation
- **Suppositories**
 - ✓ Medication that is inserted into the rectum, vagina, or urethra for absorption into the body
 - ✓

D. Other dosage form

- **Transdermal delivery systems**

- ✓ Elicits clinical responses by carrying medications across the skin barrier to the bloodstream.
- ✓ Transdermal gels and patches that are used in companion animals.
- ✓ Unique to veterinary medicine, such as spot-on, and pour-on developed for the control of parasites.

- **Inhalants**

- ✓ Drugs that produce mind-altering effects when their vapors or fumes are inhaled. They typically consist of volatile substances that produce chemical vapors when they are inhaled.

- **Infusion drugs**

- ✓ Medications that are administered intravenously (through a vein) over a period of time.

1.4.2. Dose calculations

- ✓ Dose can be given commonly in mg/kg body weight.
- ✓ The body weight of the animal should be estimate To calculate the total amount of a drug to be administrated in to a given animal

I. Percentage solutions

Many drugs are supplied in a concentrated form, and you may be required to either reconstitute a powdered drug with water to give the desired concentration, or to dilute a liquid concentrate. Some injectable drugs are supplied at a given concentration with a dose rate in mg/kg. Remember that:

- 1kg = 1000g
- 1mg = 1000mcg
- 1g = 1000mg
- 1L = 1000mls

To convert the concentration of a solution expressed in percentages into mg/ml simply multiply the percentage figure by 10.

- % solution = weight in grams x100 divided by the volume of solution in mls.
- Weight in grams = volume of solution in ml x % solution divided by 100.

Example –a bitch is admitted. She weighs 20kg and requires a daily injection of a 7.5% drug in solution. The dose is 10mg/kg. Calculate the volume of the dose to be administered.

- A 7.5% solution = 75mg in 1ml.
- Dose rate = 10mg/kg.
- Dose rate x body weight (10 x 20) = 200mg.
- Mg required divided by mg in solution (200 divided by 75) = 2.67mls/day.

II. Calculating Drug Dosages

- Accurate calculation of the dose is required for proper use of any pharmaceuticals in wildlife rehabilitation.
- Drug dosage calculation is relatively easy and requires very basic knowledge of mathematics.
- To determine the correct dosage of the drug, you will need to know the body weight of the animal, the recommended dosage of the drug in question for the species you are treating, and the concentration of the drug.
- The body weight of the animal and the recommended dosage must be expressed in the same units.
- That is, if the recommended dosage is in grams or kilograms, the animal's weight must be expressed in grams or kilograms
- The following formula is used for calculating the dose of drug to be administered to the patient:

Animal weight (kg) x dosage (mg/kg) / concentration of drug (mg/ml) = drug volume in ml

Example 1 A 75 lb deer is prescribed penicillin at a dose of 10,000 IU/lb BID.

- The concentration of the penicillin is 100,000 IU /cc.
- **Dose** = 75 lb X 10,000 IU /lb / 100,000 IU /cc
- **Dose**= 7.5 cc
- **Daily dose** is 7.5 cc x 2 = 15 cc

Example 2: A 3 lb. Great horned owl is prescribed Baytril injections at the rate of 10 mg/kg SID.

- The Baytril concentration is 22.7 mg/ml.
- First convert owl's weight to kg:
- 3 lb X 1 kg/2.2lb = 1.36 kg

- Dose = $1.36 \text{ kg} \times 10 \text{ mg/kg} / 22.7 \text{ mg/ml}$
- Dose = 0.60ml
- Daily dose is 0.6 ml

III. Administration of Tablets

- Tablets and capsules are often supplied in various strengths.
- They may be classified in micrograms (mcg), milligrams (mg) or grams (g).
- Drug doses may be calculated by either body weight (commonly) or by body surface area.

Example:

- The dog's body weight in kilograms. There are 2.2 pounds in 1 kg. If the dog is 32 pounds, then divide 32 by 2.2 pounds to get 14.5 kg
- If you are giving an antibiotic injection and the dose is 5 mg/kg, you will want to multiply the dose 5 mg/kg X 14.5 kg (the kg cancel out) to give you 72.5 mg total dose for the patient.
- Now, you also need to know how many mg are in 1 mL of the antibiotic solution so you can give the right dose of medication.
- Note that this medication gives you a percent concentration, so you will have to convert this percent into milligrams per milliliter (mg/mL).
- Use this formula for these types of questions: "mg/ml=%X10".
- Using this formula, multiply the percentage by 10. So $2.27\% \times 10 = 22.7 \text{ mg/ml}$.
- Side note- (if the percent concentration was 5% it would be 50mg/mL, if it was 10% it would be 100mg/mL, and so on.)
- If the solution is 22.7 mg/mL, how many ml will you give your patient? Take your 72.5 mg total dose needed divided by the strength of 22.7 mg/mL (mg cancel out) and you get 3.2 ml. You will give your patient 3.2 mls (the equivalent of 72.5 mg) of a 22.7 mg/mL Baytril solution IV over 20 minutes.

IV. Preparation of 10% formalin from 40% formaldehyde

- Use formula $C1V1=C2V2$

$V1$ = Required volume, $C2$ = Required concentration, $C1$ = Available percent, $V2$ = given volume

Eg.1. what volume of formaldehyde is needed to prepare 100ml of 10% formalin from 40% formaldehyde solution?

$$C_2V_2/C_1=V_1$$

V_1 = required volume, C_2 = 10% -required percent, C_1 = 40% -available percent, V_2 = 100ml
 V_{H_2O} –Volume of distilled H_2O that should be added.

$$C_1V_1 = C_2V_2 \quad V_1 = C_2V_2/C_1 \quad V_1 = 10\% \times 100\text{ml} / 40\% = 25\text{ml}$$

- Therefore 25 ml of 40% formaldehyde is needed to prepare 100 ml of 10% formalin solution.

V. Calculation of drop factor

- Drop Factor = 60 drops per ml. = $83.833 \approx 84$ drops/minutes. The formula to calculate how many hours will it take for the IV to complete before it runs out is: Time (hours) = Volume (mL) Drip Rate (mL/hour)

VI. Poultry drug dosage calculations

- Calculating poultry drug dosages requires consideration of factors such as the type of drug, the weight of the poultry, the desired therapeutic effect, and the recommended dosage guidelines provided by the drug manufacturer or a veterinarian.
- General considerations for poultry drug dosage calculations:
 - ✓ Determine the weight of the poultry: Accurate dosage calculations require knowing the weight of the individual bird or the average weight of the flock.
 - ✓ Check the drug label or package insert: The drug label or package insert typically provides dosage guidelines based on the weight of the poultry or the concentration of the drug to be administered. Follow the instructions provided by the manufacturer or consult with a veterinarian for proper dosing.
 - ✓ Calculate the dosage: Based on the drug concentration and the weight of the poultry, calculate the appropriate dosage. This may involve converting between different units of measurement if necessary.
 - ✓ Consider route of administration: Some drugs are administered orally, while others may be given via injection, water medication, or feed medication. Ensure that you understand the appropriate method of administration for the specific drug.

- ✓ Adjust for dilution or mixing: If the drug needs to be diluted or mixed with a carrier, follow the instructions for proper preparation. Ensure accurate measurements and thorough mixing to achieve the desired concentration.
- ✓ E. g. Calculate the dosage of amprolium to administer to poultry via water?
 - ✚ Consider the concentration of the drug, the desired dosage, and the total water volume consumed by the birds.
 - ✚ Determine the concentration of amprolium (Check concentration (mg/ml)).
 - ✚ Determine the desired dosage: vary depending on the specific species, age, weight, and condition being treated.
 - ✚ Calculate the total water volume consumed: Estimate the average daily water consumption of your poultry flock. This can vary based on factors such as temperature, bird size, and activity level. For example, if your flock consumes 10 liters (10,000 ml) of water per day, use this value for the calculation.
 - ✚ Calculate the required amount of amprolium: Multiply the desired dosage (in mg) by the total water volume consumed (in ml) by the birds per day. This will give you the total amount of amprolium required for the flock in milligrams (mg).
 - ✚ Convert the dosage to the appropriate volume: Divide the total amount of amprolium required (in mg) by the concentration of amprolium (mg/ml) to determine the volume of amprolium solution needed.
 - ✚ Prepare the amprolium solution: Dilute the calculated volume of amprolium in the appropriate amount of water to achieve the desired concentration.

1.4.3. Routes of drug administration

- It is the path by which a drug, fluid, poison, or other substance is taken into the body.

- Most of the drugs can be administered by different routes.
- Drug- and patient-related factors determine the selection of routes for drug administration.

The factors are:

- ✓ Physical and chemical properties of drugs (solid/ liquid/ gas; solubility, stability, pH, irritancy). Emergency/routine use,
- ✓ Site of action of the drug—local or systemic,
- ✓ Rate and extent of absorption of the drug from different routes,
- ✓ Condition of the patient (unconscious, vomiting, diarrhea),
- ✓ Age of the patient,
- ✓ Effect of gastric pH, digestive enzymes and first-pass metabolism,
- ✓ Rapidity with which the response is desired (routine treatment or emergency),
- ✓ Accuracy of dosage required (IV and inhalational can provide fine tuning)

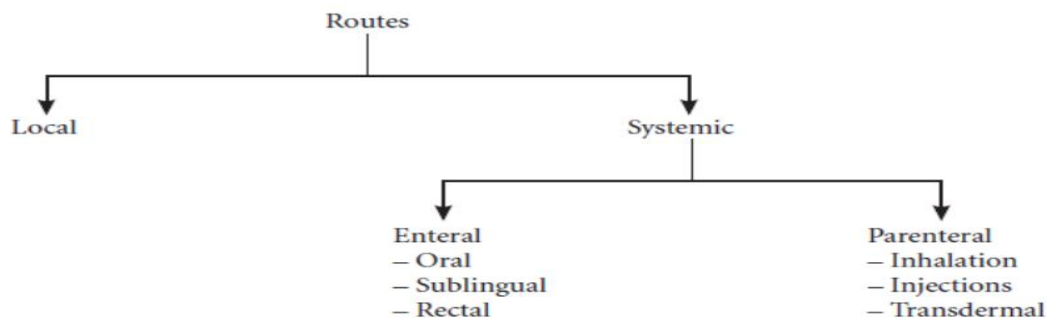


Figure 1.2: Classification of routes of drug administrations

A. Local Routes of drug administrations

- It is the simplest mode of administration of a drug at the site where the desired action is required.
 - Systemic side effects are minimal.
- I. **Topical:** Drug is applied to the skin or mucous membrane at various sites for local action.
- Skin: Ointment, cream, lotion, paste, powder, dressings, spray etc. e.g. clotrimazole
 - Mucous membranes:
 - Mouth & Pharynx – Paint, mouth wash, gargles etc.
 - Eye, ear, nose – Drops, ointment, nasal spray e.g. gentamicin eye/ear drops.
 - GI tract – As non-absorbable drugs given orally, e.g. Mg (OH)₂, sucralfate, neomycin etc.
 - Bronchi and lungs

- As inhalation, e.g. salbutamol
- Gases, volatile liquids and solids (in the form of finely divided powders) are inhaled for systemic and local effects.

Advantages

- ✓ Rapid absorption of the drug due to large surface area.
- ✓ First pass effect is avoided,
- ✓ Rapid local effects

Disadvantages

- ✓ Only few drugs can be administered,
- ✓ May produce irritation of pulmonary mucosa.
- ✓ Inconvenient procedure.
- ✓ Chances of cardio toxicity
- Urethra – As jellys e.g lidocaine; irrigating solutions.
- Rectum, Vaginal and anal canal:
 - ✓ As an enema (administration of drug into the rectum in liquid form):
 - ✚ Evacuate enema (for evacuation of bowel): For example, soap water enema— soap acts as a lubricant and water stimulates the rectum.
 - ✚ Retention enema: For example, methylprednisolone in ulcerative colitis.
 - ✚ As a suppository (administration of the drug in a solid form into the rectum), e.g. bisacodyl— for evacuation of bowels.
- Vagina – As pessaries (vaginal suppositories), vaginal tablets, inserts, cream, powders, douches.
- Anal canal – As ointment, suppositories.

Advantages

- ✓ Little first pass effect,
- ✓ Can be given in vomiting,
- ✓ Can be given in unconscious patient,
- ✓ Higher therapeutic concentrations of drug are achieved rapidly in rectum.
- ✓ For rapid evacuation of bowel, usually during gut sterilization before any surgical or radiological procedure.

Disadvantages

- ✓ Inconvenient,
- ✓ Drug absorption is slow and erratic,
- ✓ Irritation or inflammation of rectal mucosa can occur.
- **Transdermal:**
 - ✓ Can provide prolonged or controlled drug delivery.
 - ✓ Systemic absorption (Transdermal) is better with low dose, low MWt, lipid soluble drugs.

II. Systemic Routes (Enteral)

A. Enteral

- Drugs administered by this route enter blood and produce systemic effects.

Enteral Routes It includes:-

- **Oral route:**
 - ✓ Oldest and commonest mode of drug administration,
 - ✓ Safer and more convenient,
 - ✓ Medicament need not be sterile,
 - ✓ Solid dosage forms: Powders, tablets, boluses, capsules
 - ✓ Liquid dosage forms: Elixirs, syrups, emulsions, mixtures etc.
 - ✓ Advantages:
 - ✚ Convenient for repeated and prolonged use.
 - ✚ Portable, safe, no pain, can be self-administered.,
 - ✚ Cheap - no need to sterilize (but must be hygienic of course),
 - ✚ Variety of dosage forms available,
 - ✚ Fast release tablets, capsules, enteric coated, layered tablets, slow release, suspensions, mixtures,
 - ✓ Disadvantages:
 - ✚ Action is slower – not suitable for emergencies,
 - ✚ Unpalatable drugs are difficult to administer,
 - ✚ May cause nausea (tendency to vomit),
 - ✚ Can't be used for uncooperative, unconscious or vomiting patient,
 - ✚ Certain drugs are not absorbed by oral route (e.g. streptomycin)
 - ✚ Some drugs are destroyed by gastric juices (e.g. Penicillin G),

- ✚ Sometimes inefficient: high dose or low solubility drugs may suffer poor availability, only part of the dose may be absorbed,
- ✚ First-pass effect: drugs absorbed orally are transported to the general circulation via the liver. Thus drugs which are extensively metabolized will be metabolized in the liver during absorption.
- ✚ Food and GIT motility can effect drug absorption,
- ✚ Local effect: Antibiotics may kill normal gut flora and allow overgrowth of fungal varieties. Thus, antifungal agent may be included with an antibiotic.

- **Sublingual or buccal route:**

- ✓ Taken as smaller tablets which are held in the mouth or under the tongue.
- ✓ Advantages:-
 - ✚ Quick onset of action,
 - ✚ Action can be terminated by spitting out the tablet,
 - ✚ By passes first-pass metabolism, and
 - ✚ Self-administration is possible.
 - ✚ Not suitable for bitter tasting and unpalatable drug,
 - ✚ Not suitable for Irritant and lipid-insoluble drugs, cannot give to unconscious patient,
 - ✚ Large quantities cannot be given, and
 - ✚ Cannot be given in severe vomiting.

- ✓ Disadvantages:-

- **Rectal Route:**

- ✓ Drugs can be given in the form of solid or liquid.
 - ✚ **Suppository:** It can be used for local (topical) effect as well as systemic effect, e.g. indomethacin for rheumatoid arthritis.
 - ✚ **Enema:** Retention enema can be used for local effect as well as systemic effect. The drug is absorbed through rectal mucous membrane and produces systemic effect, e.g. diazepam for status epilepticus in children.
- ✓ Advantages
 - ✚ Little first pass effect,
 - ✚ Can be given in vomiting,
 - ✚ Can be given in unconscious patient,

- ✚ Higher therapeutic concentrations of drug are achieved rapidly in rectum,
- ✚ For rapid evacuation of bowel, usually during gut sterilization before any surgical or radiological procedure.
- ✓ Disadvantages
 - ✚ Inconvenient, not well accepted, May be some discomfort,
 - ✚ Drug absorption is slow and erratic,
 - ✚ Irritation or inflammation of rectal mucosa can occur

B. Parenteral

- Routes of administration other than enteral route are called parenteral routes.
- Advantages
 - ✓ Onset of action of drugs is faster; hence it is suitable for emergency,
 - ✓ Useful in: - Unconscious patient,
 - ✓ Uncooperative and unreliable patients,
 - ✓ Patients with vomiting and diarrhea.
 - ✓ It is suitable for:
 - ✚ Irritant drugs,
 - ✚ Drugs with high first-pass metabolism,
 - ✚ Drugs not absorbed orally,
 - ✚ Drugs destroyed by digestive juices.
- Disadvantage:
 - ✓ Require aseptic conditions,
 - ✓ Preparations should be sterile and is expensive,
 - ✓ Requires invasive techniques that are painful,
 - ✓ Cannot be usually self-administered,
 - ✓ Can cause local tissue injury to nerves, vessels, etc.

I. Intravenous (IV)

- Drugs are administered as:

- ✓ Bolus: Single, relatively large dose of a drug injected rapidly or slowly as a single unit into a vein. Example, IV ranitidine in bleeding peptic ulcer
- ✓ Slow intravenous injection: Example, IV morphine in myocardial infarction
- ✓ Intravenous infusion: Example, fluids infused intravenously in dehydration.
- Advantages
 - ✓ Bioavailability is 100%.
 - ✓ Quick onset of action;
 - ✓ Route of choice in emergency, e.g. IV diazepam to control convulsions in status epilepticus
 - ✓ Large volume of fluid can be administered, e.g. IV fluids in patients with severe dehydration.
 - ✓ Highly irritant drugs, e.g. anticancer drugs can be given because they get diluted in blood.
- Disadvantages
 - ✓ Once the drug is injected, its action cannot be halted,
 - ✓ Local irritation may cause phlebitis,
 - ✓ Self-medication is not possible,
 - ✓ Strict aseptic conditions are needed,
 - ✓ Extravasation of some drugs can cause injury, necrosis and sloughing of tissues.
- Precautions
 - ✓ Drug should usually be injected slowly
 - ✓ Before injecting, make sure that the tip of the needle is in the vein.
- Sites for venipuncture in different species
 - ✓ **Cattle, sheep and goat** – At any point along the whole length of the jugular vein in the jugular furrow, on the ventero-lateral aspect of the neck on either side
 - ✓ **Horse** – External jugular vein in the jugular furrow only in the cranial part of the neck.
 - ✓ **Pig** – Auricular vein
 - ✓ **Dog** – saphenous vein on the medial aspect of the leg or recurrent tarsal vein on the dorsal aspect of the leg.

II. Subcutaneous (S.C.) route

- The drug is injected into the subcutaneous tissues of the thigh, abdomen and arm, e.g. adrenaline, insulin, etc.
- **Advantages:**
 - ✓ Actions of the drugs are sustained and uniform,
 - ✓ Drugs can be given in presence of vomiting and diarrhea,
 - ✓ Drugs can be given to unconscious patients,
 - ✓ First pass effect is avoided,
 - ✓ Drugs that are not absorbed from G.I.T can be given,
 - ✓ Self-administration is possible (e.g. insulin),
 - ✓ Depot preparations can be inserted into the subcutaneous tissue, e.g. norplant for contraception.
- **Disadvantages**
 - ✓ Only non-irritant drugs can be given otherwise severe irritation, pain and necrosis of subcutaneous tissues can occur,
 - ✓ Absorption of the drugs is slow than I/M injection,
 - ✓ Expensive,
 - ✓ Danger of infection, if proper sterilization techniques are not used,
 - ✓ Large volumes of drug cannot be given.
- **Sites for subcutaneous injection in different species**
 - ✓ **Cattle, sheep and goat** – Fold of flank extending from the caudoventral abdominal wall to the craniomedial aspect of the thigh near stifle joint or loose skin on the lateral aspect of the neck.
 - ✓ **Horse** – Loose skin on the lateral aspect of the neck. Normally subcutaneous injection is not preferred in horses.
 - ✓ **Dog** – Fold of the flank

III. Intramuscular (IM) route

- The drug is injected deep in the belly of a large skeletal muscle.
- The muscles that are usually used are deltoid, triceps, Gluteus, Maximus, rectus, femurs depending on the specie of animal.
- The muscle is less richly supplied with sensory nerves; hence injecting a drug is less painful.
- Absorption of drug slows especially in high fat deposition area.

- Intramuscular injections are given at an angle of 45 degrees in Gluteus muscle region of bovine.
- Advantages
 - ✓ Rate of absorption is uniform.
 - ✓ Rapid onset of action.
 - ✓ Irritant substances can be given.
 - ✓ Drugs can be given to unconscious patients.
 - ✓ Accuracy of dosage is ensured.
 - ✓ Useful in emergency situations.
 - ✓ First pass effect is avoided.
 - ✓ Drugs producing gastric irritation can be given.
 - ✓ Drugs that are not absorbed from G.I.T can be given.
- Disadvantages
 - ✓ Local pain and abscess formation.
 - ✓ Technical person is needed, self-administration is difficult.
 - ✓ Danger of infection, if proper sterilization techniques are not used.
 - ✓ Chances of nerve damage
- **Sites for intramuscular injection in different species**
 - ✓ **Cattle, sheep and goat**
 - ✚ **Hind limb**
 - » Gluteal region covered by gluteal muscles (Preferred in bovines).
 - » Posterior aspect of thigh between the semimembranosus and semitendinosus (Preferred in sheep and goat).
 - ✚ **Neck**
 - » In the heavy muscles on the caudo-dorsal aspect of the neck (preferred only in well-built animals).
 - » Care should be taken to avoid vertebral column and the dorsal branch of the XI cranial nerve.
 - ✚ **Hump**– In indigenous cattle with hump, hump is also preferred to administer intramuscularly.
 - ✓ **Horse** – Neck region and brisket region (pectoral muscles)
 - ✓ **Dog** – Hind limb between the semimembranosus and semitendinosus.

IV. Intra-peritoneal Route

- Indication: Colon and Ovarian
- Peritoneal space has much surface area; may not be reached by IV chemo
- Catheters used: implanted port
- Chemotherapy agents used: Cisplatin, Taxol
- Advantages: less systemic side effects
- Disadvantages: infection, pain.

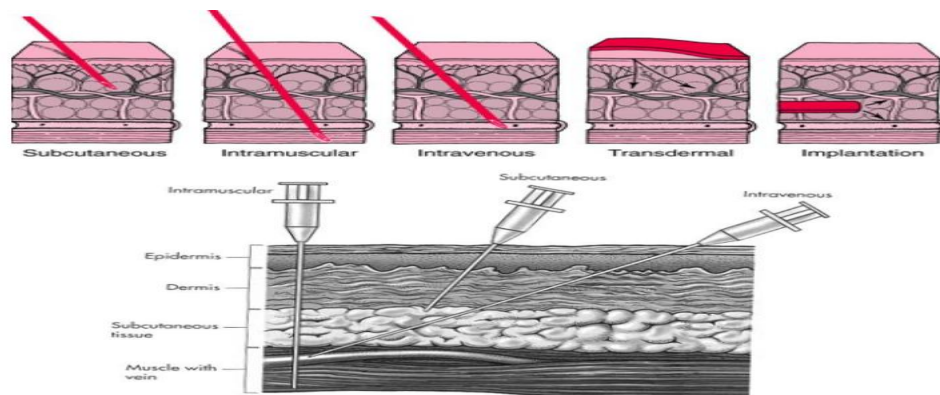


Figure 1.2: Parenteral of drug administrations

- **Note: Order of absorption through various routes:**
 - ✓ i.v. > inhalation > i.m. > i.p. > s.c. > ora

1.5. Drug prescription and pharmacological abbreviations

1.5.1. Drug prescription

- A **prescription** is defined as a health care program implemented by a physician in the form of instruction that govern the plan care for an individual patient.
- Prescription writing is a crucial task and suggests prescriber's responsibility towards the clinical care and the safe monitoring of the patient thus also carries legal implication.
- A **prescription drug** is a pharmaceutical drug that is permitted to be dispensed only to those with a medical prescription.
- There are four aims that a prescriber should try to achieve, they are:
 - ✓ To maximize effectiveness,
 - ✓ Minimize costs, and
 - ✓ Minimize risks,
 - ✓ Respect the patient's choices.

1.5.2. Parts of the Prescription

- **Prescription** should state what is to be given to whom and by whom prescribed, and give instructions on how much should be taken how often, by what route and for how long or total quantity to be supplied.
- The following are parts of Prescription:
 - ✓ Date
 - ✓ Address of doctor
 - ✓ Superscription {Symbol (Rx)}
 - ✓ Inscription or the name and dose of medication prescribed
 - ✓ Subscription or Dispensing direction to Pharmacist
 - ✓ Signature or Instructions for Patient
 - ✓ Signature of doctor

1.5.3. Pharmacological abbreviation

Table 1.2: Abbreviations Describing Route of Administration

Abbreviation	Meaning	Example
PO	By mouth (orally)	Give 1 tab PO
SC	Subcutaneously	Give 1 ml SQ
IM	Intramuscular	Inject 1 ml IM
IV	Intravenous	Inject 1 ml IV
IP	Intra-peritoneal	Inject 1 ml 20 ml IP
Inj.	Injection	
OD	Right eye	Instill 1 drop OD
OS	Left eye	Instill 1 drop OS
OU	Both eyes	Instill 1 drop OU
AD	Right ear	Apply small amount AD
AS(Sometimes AL)	Left ear	Apply small amount AS
AU	Both ears	Apply small amount AU
ID	Intradermal	Inject 0.2 ml ID
w.o	Without	
a.c	Before meals	
aq	Aqua	water
ad lib.	ad libitum	use as much as one desires; freely
bis	Bis	twice
b.i.d.	bis in die	twice daily
q.o.d	quaque altera die	<i>Every Other Day</i>
s.i.d.	Single in die	Once a day
q.i.d	quater in die	Four times each day

bol.	Bolus	as a large single dose (usually IV)
Haust		drench
Gtt(s)		drop(s)

Table 1.3: Abbreviations describing quantity or dose

Abbreviation	Unit of Measure	Equivalent To
Lb (or lbs)	pound(s)	pound(s) 16 oz
mg	Milligram	0 .001 gram
g	Gram	0 .001 kg
kg	Kilogram	2.2 lb
U	Unit	Often short for international unit
U (of blood)	Unit of blood	The quantity of blood (plus anticoagulant) collected from a blood donor. For blood components, this is the volume of that component that would be present in a unit of blood. Typically, the volume is 250 ml of whole blood for dogs and 50 ml for cats.
IU (or UI)	International unit	Quantity varies for different drugs or vitamins. This is an agreed upon measure of the amount of a vitamin or drug causes a certain effect.
gr	Grain	64.7989 mg
mEq	Milliequivalent	One-thousandth of a molar equivalent of a chemical
ml	Milliliter	0 .001 liter
oz	Ounce	1/16 of a pound (30 grams
qs	Quantity sufficient	
tbsp	Tablespoon	15 ml
tsp	Teaspoon	5 ml
dr	Dram (drachm)	1/16 of an ounce (28.8 grams) A fluid dram is 1/8 of a fluid ounce
cc	Cubic centimeter	1 ml

Table 1.4: Other Abbreviations

Abbreviation	Meaning
Sig	Write on label
Stat	Immediately
Susp	Suspension
Tab	Tablet
Syr	Syrup
Cap	Capsule

Amp	Ampule
DAW	Dispense as written

1.6.The fate and mechanisms of actions of drugs

- Antibacterial targets a variety of unique bacterial features can be targeted by antibiotics w/o harming host cells:
- Antibiotics work only on bacteria and are described by their spectrum of action:
 - ✓ Narrow-spectrum: Work only on either gram-positive or gram-negative bacteria (not both)
 - ✓ Broad-spectrum: work on both gram-positive and gram-negative bacteria (but not necessarily all)
- Antibiotics can be classified as bactericidal or bacteriostatic
 - ✓ Bactericidal kill the bacteria
 - ✓ Bacteriostatic inhibit the growth or replication of bacteria

1.6.1. Mechanisms of actions of Antibiotics

A. Cell wall synthesis Inhibiter

- Irreversibly inhibit enzymes involved in the final steps of cell wall synthesis
- Drug binds to a series of enzymes involved in the final stage of cell wall synthesis, competitively inhibit enzymatic activity.
- Leads to the formation of defective cell walls that are osmotically unstable and cell death usually results from lysis
- Some bacteria produce b-lactamase- enzyme that breaks the critical b-lactam ring
- **Examples of Cell wall synthesis Inhibiter are as follows:**

I. Beta-Lactam Drugs: b-lactam ring is similar in structure to normal substrate of enzyme

- **Penicillin:** Most widely effective and the least toxic drugs
 - ✓ Mechanism of action (MoA):
 - ✚ Have b-lactam structure that interferes with bacterial cell wall synthesis by binding to Penicillin binding proteins (PBPs) of the bacterial enzyme which inhibition of transpeptidase

- ✚ Only effective against rapidly growing organisms that synthesize a peptidoglycan cell wall
- ✚ Identified by the *-cillin* ending in the drug name
- ✓ Pharmacokinetics (PK):
 - ✚ Spectrum of activity depends on the type of penicillin
 - ✚ Narrow-spectrum penicillin : For gram +ve bacteria
 - ✚ Broader/extended-spectrum penicillin are semi-synthetic used for both Gram +ve and G-ve bacteria
 - ✚ E.g. Amoxicillin, ampicillin, carbenicillin, ticarcillin, and methicillin
 - ✚ b-lactamase resistant penicillin E.g. Methicillin, oxacillin, dicloxacillin, cloxacillin, and floxacillin
- ✓ Adverse drug reaction (ADR):
 - ✚ Hypersensitivity
 - ✚ Diarrhea- due to disruption of normal flora
 - ✚ Nephritis- acute interstitial nephritis by Methicillin
 - ✚ Neurotoxicity- irritates neural tissue, aggravate seizure
- **Cephalosporin**
 - ✓ PK:
 - ✚ Semi-synthetic, broad-spectrum antibiotics
 - ✚ Closely related both structurally and functionally to the penicillins
 - ✚ Have the beta-lactam ring
 - ✚ Can be identified by the *ceph-* or *cef-* prefix in the drug name
 - ✚ In general, as the number of the generation increases, the spectrum of activity broadens (but becomes less effective against gram-positive bacteria).
 - ✓ **Antibacterial activity:**
 - ✚ Classified into four generations- 1st, 2nd, 3rd & 4th generation, based largely on their bacterial susceptibility patterns and resistance to b-lactamase
 - ✓ MoA: The same mode of action as penicillins
 - ✓ Indications: Approved for Veterinary Medicines: cephalexin, Ceftiofur, cefotaxime, cephalixin etc (Cefquinome-4th gen)

✓ ADRs: Allergic effects similar to penicillins

- **Glycopeptides:** The most common glycopeptide antibiotics include vancomycin, teicoplanin, telavancin, bleomycin, ramoplanin, and decaplanin.

✓ **Vancomycin**

MoA: inhibits synthesis of bacterial cell wall phospholipids & peptidoglycan polymerization

Pk: The drug exerts bactericidal effects to a broad spectrum of microorganisms.

Indications: Effective against many gram-positive bacteria; MDR (Multidrug resistance) organisms e.g., MRSA (Methicillin resistant Staph. Aureus) & enterococci

B. Cell Membrane Disrupting Agents

- Polypeptides

✓ Polymyxin B and E, Bacitracin

✓ Polypeptide antibiotics are thought to act by permeabilizing the bacterial cell membrane

✚ **Polymyxin B**

- » Works by attacking the cell membrane of bacteria
- » Binds to membrane of G-ve bacteria and alters permeability. This leads to leakage of cellular contents and cell death
- » These drugs also bind to eukaryotic cells to some extent, which limits their use to topical applications
- » Is a narrow-spectrum, G-ve antibiotic
- » Not absorbed when taken orally or applied topically
- » Used as an ointment or wet dressing

- **Bacitracin**

✓ Isolated from *Bacillus subtilis*

✓ Inhibits the early stage in peptidoglycan synthesis, i.e. disrupts the bacterial cell wall

✓ is effective against G+ve bacteria

✓ Forms complex with mammalian cell membrane lipids

✓ Poorly absorbed from orally

- ✓ Used topically (skin, mucous membranes, eyes) and as a feed additive?
- ✓ Because of its potential nephrotoxicity with systemic use

C. Protein Synthesis Inhibitors

I. Aminoglycosides

- Amikacin, gentamycin, neomycin, streptomycin, tobramycin
- Streptomycin first drug to be discovered among members but **toxic**.
- **Gentamicin, Neomycin, Tobramycin and Amikacin= less toxic**
- **MoA:** Irreversibly binding to 30S subunit of rRNA causing it to distort and malfunction; blocks initiation of translation (interfering with mRNA) of bacteria)
- **Antibacterial activity:** Broad spectrum of activity, used for gram-ve bacteria
- **Pk:**
 - ✓ Excellent water, but poor lipid, solubility
 - ✓ Not absorbed well from the GI tract, so are given parenterally
 - ✓ Stable over a wide ranges of pH and temperatures
 - ✓ Inadequate concentration in CSF
 - ✓ High concentration in renal cortex & inner ear
 - ✓ All of them cross the fetal membranes
 - ✓ May be recognized by *-micin* or *-mycin* ending in drug name
- **SE:**
 - ✓ Potentially nephrotoxic when given systemically and ototoxic due to binding to proximal tubules & inhibition of mitochondrial functions
 - ✓ Neuromuscular paralysis
 - ✓ Allergic rxns- contact dermatitis for topically applied

II. Tetracycline

- First generation members include Tetracycline, Chlortetracycline, Oxytetracycline and Demeclocycline. obtained by biosynthesis
- Second generation Members such as Doxycycline, Lymecycline, Meclo cycline, Methacycline, Minocycline, and Rolitetracycline (because they are derivatives of semi-synthesis.

- Those obtained from total synthesis such as Tigecycline are considered to be Third generation.
- **MoA:** Bind reversibly to 30S subunit of bacterial ribosomes, thereby blocking attachment of tRNA to mRNA-ribosome complex at receptor site. Interfere with bacterial protein synthesis
- **Antibacterial spectrum:** Broad
- **Indications:** Wide variety of bacteria, intracellular organism: Mycoplasma, Chlamydia & Rickettsia
- PK:
 - ✓ Can bind to divalent ions: calcium and be deposited in growing bones and teeth,
 - ✓ Complex formation (chelation) limits absorption
 - ✓ Adequate penetration into most body fluids
 - ✓ All of them cross the placenta barrier
 - ✓ Metabolized and conjugated to form soluble glucuronides
 - ✓ Parent &/or metabolites are secreted into the bile and most undergo enterohepatic circulation
 - ✓ Are recognized by *-cycline* ending in drug name
- **SE:** GIT signs, hepatotoxicity, and super infection. E.g. Tetracycline, oxytetracycline, chlortetracycline, doxycycline and minocycline

III. Macrolides

- Azithromycin, clarithromycin, Erythromycin, Tilimicosin, Tylosin, Gamithromycin, Tulathromycin, Tildipirosin.
- **MoA:-** Inhibits protein synthesis by irreversibly binding to 50S subunit of bacterial ribosome
- **Antibacterial spectrum:** a broad-spectrum of activity- mainly Gram+ve
- **Indications:** Used to treat penicillin-resistant infections or in animals that have allergic reactions to penicillin
 - ✓ Bacteriostatic at therapeutic concentration, but can be slowly bactericidal, especially against *Streptococci*
- PK:
 - ✓ Weak bases that are poorly absorbed

- ✓ Has a large molecular structure
- ✓ Variable distribution between the drugs
- ✓ Primarily concentrated & excreted in active form in the bile
- ✓ Optimum antibacterial pH is 8: Acidic environment: abscess, necrotic tissue or urine suppress its antibacterial activity

- **ADR:**

- ✓ Epigastric distress-oral based
- ✓ Cholestatic jaundice
- ✓ Ototoxicity

IV. Chloramphenicol

- Inhibits protein synthesis by its action on the 50S ribosome of both bacterial and mammalian cells
- Is a synthetic, has a broad-spectrum of activity that penetrates tissues and fluids well (including the eyes and CNS)
- Indication:
 - ✓ Can be used to treat various bacterial infections in animals, such as skin, wound, bone, intestinal, and respiratory infections.
 - ✓ It is effective against many types of bacteria, including those that do not need oxygen to grow (anaerobic bacteria) and those that cause
- **SE:** Toxic, Bone marrow depression that extremely limit use
 - ✓ Chloramphenicol also has some serious side effects and risks, such as blood disorders, liver or kidney damage, and reproductive problems.
 - ✓ It should be used with caution and only under the guidance of a veterinarian.
- Note: Its use is strictly controlled in food-producing animals because of
 - ✓ Drug residue effects –aplastic anemia in susceptible individuals
 - ✓ Its effect is not concentration dependent

D. Inhibit synthesis of essential metabolites

I. Sulfonamides

- MoA:
 - ✓ Synthetic analogs of PABA

- ✓ Inhibit metabolism by interfering with the synthesis of FA (needed for the growth of many organisms)
- ✓ Humans/animals obtain preformed folate in the form of folic acid as a vitamin from diet.
- ✓ However, many bacteria are impermeable to FA, and therefore, must rely on *de novo* folate synthesis.
- ✓ Bacteriostatic,
- ✓ Become bactericidal if their **concentration** is sufficiently high or if the presence of any sulfonamide concentration is accompanied by other environmental **conditions unfavorable** to bacteria
- ✓ Broad antibacterial activity: Many bacteria, Coccidia, Chlamydia, protozoa including Toxoplasma
- PK:
 - ✓ Some are designed to stay in the GI tract (sulfasalazine);
 - ✓ Some are absorbed by the GI tract and penetrate tissues
 - ✓ Acetylation in the liver and lungs is the main route of metabolism
 - ✓ Metabolites excreted via the kidneys
 - ✓ May be **potentiated** with Trimethoprim or ormetoprim; reduce the MIC needed of both drugs & therefore reduce SE.
- **Indications:**
 - ✓ Urinary, CNS, joint, respiratory, prostate infections
 - ✚ **Trimethoprim** distribute to most tissues of the body (lipophilic) & concentrates in tissues with greater acidity than plasma (e.g., prostate)
 - ✚ **Sulfasalazine** (Salazopyrin)
 - ✓ Used to treat IBD (inflammatory bowel disease) as one of its metabolites has a local anti-inflammatory effect in the colon
- SE: crystalluria, skin rashes: E.g., Sulfadimidine, sulfadiazine/ trimethoprim, sulfadimethoxine, and sulfadimethoxine/ ormetoprim

II. Cotrimoxazole (Trimethoprim + sulfamethoxazole)

- Greater antibacterial activity than equivalent quantities of either drug used alone

- Synergistic activity
- MoA: Inhibits two sequential steps in the synthesis of THFA
 - ✓ SMZ inhibits the incorporation of PABA into DHFA precursors
 - ✓ TMP prevents reduction of DHF to THF
- **Antibacterial activity:** Broader spectrum than the sulfa drugs
- **SE:** Dermatologic rxns, GIT problems, hematologic (Megaloblastic anemia, leukopenia)

E. Nucleic Acid synthesis Inhibitors

- Target enzymes required for nucleic acid synthesis

I. Fluoroquinolones

- MoA:
 - ✓ Inhibit enzymes (DNA gyrase/topoisomerase II) & Topoisomerase IV that maintain the supercoiling of closed circular DNA
 - ✓ Antibiotics with fluorine bound to the quinolone base, which increases the drug's potency, spectrum of activity, and absorption
- Antimicrobial activity:
 - ✓ Have a broad-spectrum bactericidal activity.
 - ✓ Gram -ve & +ve aerobic bacteria & mycoplasmas
 - ✓ Not active against anaerobes, except the 4th generation
 - ✚ 1st gen: Nalidixic acid:- G-ve bacteria
 - ✚ 2nd gen: Ciprofloxacin, norfloxacin, ofloxacin, Lomefloxacin, Enoxacin:- G-ve & some G+ve cocci
 - ✚ 3rd gen: Levo-/Spar-floxacin- expanded G-ve & G+ve
 - ✚ 4th gen: Moxi-/Gati-/Trova-floxacin – anaerobes, G-ve, G+ve
- PK:
 - ✓ Ca^{2+} & other divalent affects absorption from GIT
 - ✓ Well distribute into tissues & body fluids
 - ✓ Excreted via kidneys, but Moxifloxacin, primarily by the liver
 - ✓ IV preparation: Cipro & levo- floxacin
- **SE:**

- ✓ GIT (NVD), CNS effects (headache, dizziness), phototoxicity, connective tissue problems (cartilage damage in growing animals,
- ✓ Can be recognized by *-floxacin* ending in drug name
- ✓ E.g., enrofloxacin, orbifloxacin, difloxacin, ibafloxacin, danfloxacin- approved for veterinary use
- ✚ Others: Ciprofloxacin, marbofloxacin, and sarafloxacin
- ✚ **Ibafloxacin**- a new drug developed exclusively for veterinary use

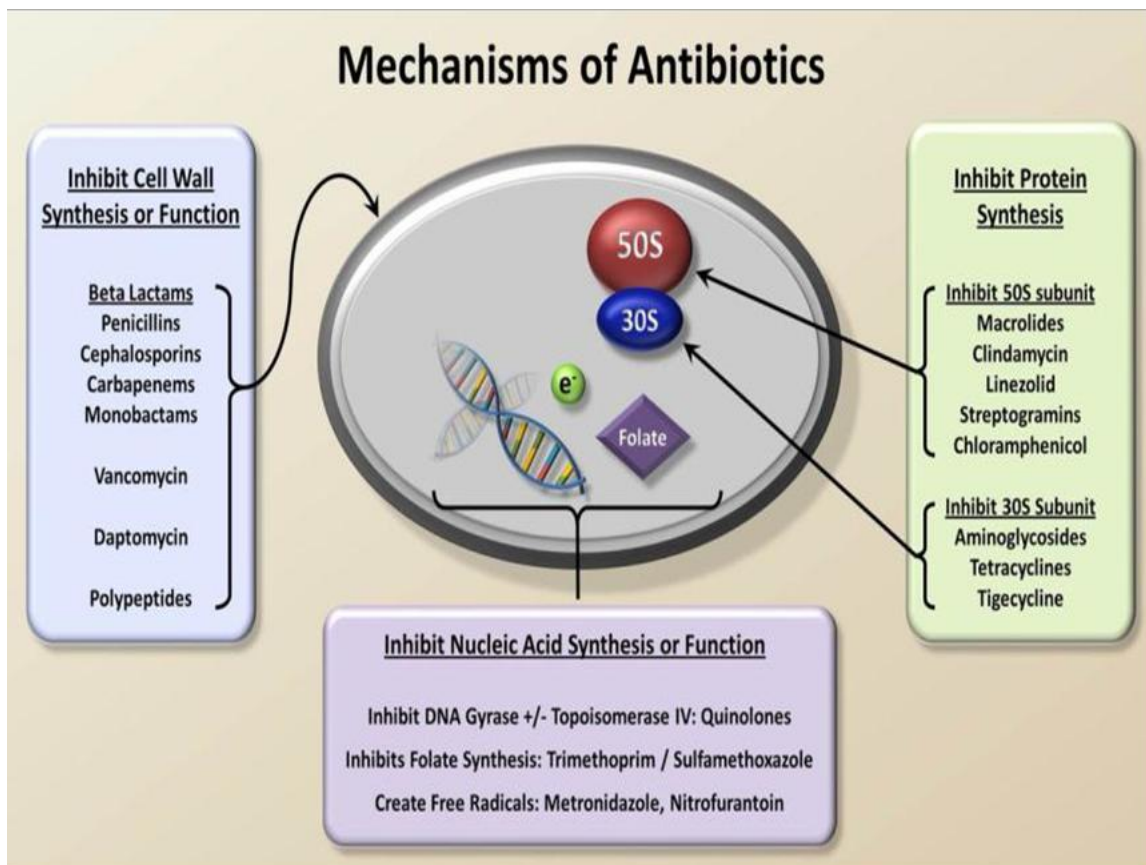


Figure 1.4: Mechanism action of antibiotics

1.6.2. Mechanism action of antiphrastic drugs

- Include Anti-helminthes, Antiprotozoal, Ectoparasiticides

I. Anthelmintic Drugs

- Includes: Anti-nematode drugs, Anti-cestodes, Anti-trematodes

A. Anti-nematodes drugs

- **Imidazothiazoles**

- ✓ Work by stimulating (agonist) the nematode's cholinergic nervous system, leading to paralysis of the parasite
- ✓ Effective against ascarids, Strongyles (Heamonchus, ostertagia, cooperia, nematodirus, etc), whipworms, and hookworms
- ✓ An example is levamisole, tetramisole.....
- ✓ **Levamisole:** A synthetic imidazothiazole derivative

MoA:

- » As an antiparasitic agent, it works by stimulating the nicotinic acetylcholine receptors in the muscles of worms, causing them to become paralyzed and unable to reproduce.
- » As an immunomodulator, it works by enhancing the activity and proliferation of T cells, monocytes, macrophages, and neutrophils, which are involved in fighting infections and diseases

PK:

- » Rate of absorption differs with the route of administration.
- » Fast absorption via IM or SC
- » Used with caution in patients with liver dysfunction

Indication:

- » Highly effective in eradicating Ascaris and trichostrongylus and moderately effective against species of hookworm.
- » Effective against mature and immature forms of helminths within the intestinal tract but not against migratory stages in the tissues or against ova.
- » Dogs: GI nematodes and Dirofilaricides
- » Effective against both mature & immature larval stages of lung worms

 **SE:** Signs of toxicity: salivation, defecation and respiratory distress

- **Piperazine:** Belongs to the hetrocyclic group: phenothiazine

✓ **MoA:**

- ✚ Acts by binding to the GABA receptors in the muscles of the worms, which leads to hyperpolarization of the nerve endings and flaccid paralysis.
- ✚ It also blocks the production of succinate, which is an energy source for the worms.
- ✚ The parasites, paralyzed and depleted of energy, are expelled by peristalsis

✓ **PK:**

- ✚ Usually given orally as a syrup, tablet, or powder.
- ✚ The dosage and duration of treatment depend on the type and severity of the infection, as well as the animal's weight and age.
- ✚ Used as a solvent, a chemical intermediate, and a component of some carbon capture and storage systems

✓ **Indication:**

- ✚ Effective only against ascarids (Toxocara, Toxascaris)
- ✚ No longer recommended for treatment of pinworm infection, because a 7-day course of treatment is required.
- ✚ Not useful in hookworm infection, trichuriasis, or strongyloidiasis.

✓ **SE:**

- ✚ Generally well tolerated, but some side effects may occur, such as vomiting, diarrhea, lethargy, or allergic reactions
- ✚ Should not be used in animals with liver or kidney problems, or in pregnant or lactating animals.

- **Avermectins (macrocyclic lactones)**

- ✓ Avermectins are a group of drugs that belong to the macrocyclic lactones, which are derived from a soil bacterium called Streptomyces avermitilis
- ✓ Another example in this group: moxidectin, abamectin, doramectin, milbemycin oxime, eprinomectin, selamectin

✓ **MoA:**

- ✚ Based on their ability to bind to glutamate-gated chloride channels, which are found only in invertebrates. These channels allow chloride ions to enter the nerve and

muscle cells, making them less excitable. When avermectins bind to these channels, they increase the flow of chloride ions, leading to hyperpolarization and paralysis of the cells.

- ✚ They work by blocking the nerve signals in the parasites, causing them to become paralyzed and die

✓ **PK:**

- ✚ Prototype: **ivermectin**,
 - » used for a wide variety of endo- and ectoparasites
 - » May be combined with other antiparasitic agents to broaden its spectrum of activity
- ✚ Can be administered orally, parent rally, topically, or as spot-on drops, depending on the formulation and the target parasite.
- ✚ The dosage and duration of treatment vary depending on the type and severity of the infection, as well as the animal's weight and age.
- ✚ Not effective against cestodes or trematodes

✓ **Indication:**

- ✚ Have a broad spectrum of activity against many types of nematodes and arthropods, including those that cause roundworm, pinworm, heartworm, mange, scabies, fleas, ticks, and lice. They are effective against both immature and adult stages of the parasites.
- ✚ Used for heartworm prevention

✓ **SE:**

- ✚ Generally well tolerated by animals, but some side effects may occur, such as vomiting, diarrhea, lethargy, or allergic reactions.
- ✚ Should not be used in animals with liver or kidney problems, or in pregnant or lactating animals.
- ✚ May also have negative impacts on the environment, especially on dung-inhabiting insects that play a role in nutrient cycling.

B. Anti-cestodal agents

- Used to treat infections caused by tapeworms in animals and humans. Tapeworms are a type of parasitic worm that belong to the class Cestoda
- Interfere the ability of tapeworms to maintain their location in the gut or kill the tapeworms in situ. E. g. Praziquantel, Albendazole....
- **Praziquantel**
 - ✓ Used to treat infections caused by parasitic worms, such as tapeworms and flukes, in animals and humans.
 - ✓ MoA:
 - ✚ It works by affecting the calcium channels in the worms' cells, causing them to contract and lose their grip on the intestinal wall. The worms are then either destroyed or expelled from the body
 - ✓ **Indications:**
 - ✚ Works on **adult stages of all cestode species** (also used to eliminate fleas)
 - ✚ Both human and veterinary use
 - ✚ Effective against most species of cestodes (tapeworms) and trematodes (flukes), but not against nematodes (roundworms).
 - ✓ **PK:**
 - ✚ Given orally or by injection, depending on the type of infection and the animal species.
 - ✚ The dosage and duration of treatment vary according to the type and severity of the infection. Some common uses of praziquantel in animals are:
 - ✚ In ruminants (such as cattle, sheep, and goats), praziquantel is used to treat cestodes such as *Moniezia* and *Stilesia*, and trematodes such as *Schistosoma*.
 - ✚ In horses, praziquantel is used to treat cestodes such as *Anoplocephala*1.
 - ✚ In dogs and cats, praziquantel is used to treat cestodes such as *Dipylidium*, *Taenia*, and *Echinococcus*, and trematodes such as *Paragonimus* and can be given as a tablet, an injection, or a spot-on formulation.
 - ✚ In poultry, praziquantel is used to treat cestodes such as *Raillietina*1.
 - ✚ In fish, used to treat cestodes such as *Bothriocephalus* and *Diphyllbothrium*, and trematodes such as *Clonorchis* and *Opisthorchis*.
 - ✓ **SE:**

- ✚ Generally well tolerated by animals, but some side effects may occur, such as vomiting, diarrhea, lethargy, salivation, itching, or allergic reactions.
 - ✚ It should not be used in animals that are pregnant, lactating, or hypersensitive to the drug.
 - ✚ It should also be used with caution in animals that have liver or kidney problems.
- **Albendazole:** This is a benzimidazole anthelmintic
 - ✓ **MoA:**
 - ✚ Affects the glucose uptake and metabolism of the worms, leading to their depletion of glycogen and ATP, which are the main sources of energy for the worms. Albendazole ultimately causes the death and expulsion of the worms from the body.
 - ✓ **PK:**
 - ✚ Albendazole is given orally as a tablet or a suspension
 - ✚ The dosage and duration of albendazole treatment depend on the type and severity of the infection and the animal species
 - ✓ **Induction:**
 - ✚ Ascariasis: an infection caused by the roundworm
 - ✚ Pinworm infection: an infection caused by the roundworm *Enterobius vermicularis*
 - ✚ Hookworm infection: an infection caused by the roundworms
 - ✚ Trichuriasis: an infection caused by the whipworm *Trichuris trichiura*,
 - ✚ Strongyloidiasis: an infection caused by the roundworm *Strongyloides stercoralis*
 - ✚ Taeniasis: an infection caused by the tapeworms *Taenia saginata* or *Taenia solium*,
 - ✚ Clonorchiasis: an infection caused by the liver fluke *Clonorchis sinensis*,
 - ✚ Opisthorchiasis: an infection caused by the liver fluke *Opisthorchis viverrini* or *Opisthorchis felinus*
 - ✚ Cutaneous larva migrans: an infection caused by the larval stage of various hookworms or threadworms.
 - ✚ Gnathostomiasis: an infection caused by the larval stage of various nematodes that belong to the genus *Gnathostoma*
 - ✓ **SE:**

- + Vomiting, diarrhea, anorexia, or bone marrow suppression.
- + Albendazole should not be used in pregnant or lactating animals.

- **Fenbendazole:** A drug that belongs to the class of benzimidazole anthelmintics,
 - ✓ **MoA:** Affects the glucose uptake and metabolism of the worms, leading to their depletion of glycogen and ATP, which are the main sources of energy for the worms
 - ✓ **Indications:**
 - + Roundworm infection: an infection caused by various species of nematodes, such as Ascaris, Toxocara, Ancylostoma, Necator, Trichuris, and Strongyloides
 - + Tapeworm infection: an infection caused by various species of cestodes, such as Taenia, Echinococcus, Dipylidium, and Mesocostoides
 - + Fluke infection: an infection caused by various species of trematodes, such as Fasciola, Dicrocoelium, Clonorchis, Opisthorchis, Paragonimus, and Schistosoma
 - ✓ **PK:**
 - + The dosage and duration of fenbendazole treatment depend on the type and severity of the infection and the animal species.
 - + Usually given orally as a tablet or a suspension.
 - + Limited absorption in the intestinal tract after oral administration.
 - + When consumed, not more than 10% of the medication is absorbed.
 - + Bioavailability can be increased by the presence of food in the gut.
 - + Food slows down the intestines and increases blood flow to the digestive tract, improving the absorption of the medication.
 - + Fenbendazole is mainly metabolized in the liver
 - ✓ **SE:**
 - + Vomiting, diarrhea, anorexia (loss of appetite), or bone marrow suppression (reduced production of blood cells).
 - + Should not be used in animals that are pregnant or lactating (producing milk), or have liver or kidney problems.

C. Anti-trematodals: Drugs that are used to treat infections caused by trematodes, also known as flukes, in animals and humans. Some of the common anti-trematodal drugs used in animals are:

- **Triclabendazole** : a drug that belongs to the class of benzimidazole anthelmintics,

✓ **MoA:**

- ✚ Affects the glucose uptake and metabolism of the worms, leading to their depletion of glycogen and ATP, which are the main sources of energy for the worms
- ✚ Ultimately causes the death and expulsion of the worms from the body

✓ **Indications:**

- ✚ Used to treat infections caused by parasitic worms in animals and humans.
- ✚ Used for the treatment of fascioliasis, an infection caused by liver flukes of the genera *Fasciola* and *Dicrocoelium*
- ✚ It is effective against both immature and mature stages of *Fasciola hepatica* and *Fasciola gigantica*, the most common liver flukes in animals.

✓ **PK:**

- ✚ The dosage and duration of triclabendazole treatment depend on the type and severity of the infection and the animal species.
- ✚ Usually given orally as a tablet or a suspension.
- ✚ Limited absorption in the intestinal tract after oral administration.
- ✚ When consumed, not more than 10% of the medication is absorbed.
- ✚ Bioavailability can be increased by the presence of food in the gut.
- ✚ Food slows down the intestines and increases blood flow to the digestive tract, improving the absorption of the medication.
- ✚ Mainly metabolized in the liver
- ✚ Triclabendazole and its metabolites are excreted mainly in the feces, and to a lesser extent in the urine.

✓ **SE:**

- ✚ Vomiting, diarrhea, anorexia (loss of appetite), or bone marrow suppression (reduced production of blood cells).
- ✚ It should not be used in animals that are pregnant or lactating (producing milk), or have liver or kidney problems.

- **Albendazole** Effective for all adult stages

- **Praziquantel:** Covered previously, Also effective against lung trematodes in dogs and cats

II. Mechanism of action antiprotozoal drugs

- Antiprotozoal drugs are drugs that are used to treat infections caused by protozoa, which are single-celled organisms that can cause diseases such as malaria, giardiasis, and amoebiasis.
- Antiprotozoal drugs work by different mechanisms of action, depending on the type and stage of the protozoan parasite.
- **Some of the antiprotozoal drugs:**

A. Coccidiostats

- Coccidiostats are compounds that are widely used as **feed additives** to prevent and treat coccidiosis, a contagious disease affecting mainly poultry.
- Coccidiostats are drugs that are used to prevent or treat coccidiosis, a disease caused by protozoan parasites of the genus *Eimeria* that infect the intestinal tract of poultry and other animals.
- Coccidiostats work by different mechanisms of action, depending on the type and stage of the parasite.
- **Common mechanisms of action are:**
 - ✓ **Inhibiting folic acid synthesis:** Folic acid is an essential vitamin for the growth and reproduction of protozoa. Some coccidiostats, such as sulfonamides and pyrimethamine, block the enzymes involved in the synthesis of folic acid in the protozoan cells, thus depriving them of this vital nutrient. This mechanism is effective against most species of *Eimeria*.
 - ✓ **Damaging DNA or inhibiting DNA replication:** DNA is the genetic material that carries the information for the function and structure of the protozoan cells.
 - ✓ Some coccidiostats, such as nitrofurans and quinolones, cause breaks or nicks in the DNA strands or prevent the DNA from replicating. This mechanism is effective against some species of *Eimeria*.
 - ✓ **Affecting calcium channels or energy metabolism:**

- ✚ Calcium channels are proteins that regulate the flow of calcium ions across the cell membrane, which are important for the contraction and relaxation of the protozoan cells.
- ✚ Energy metabolism is the process by which the protozoan cells produce and use energy from nutrients.
- ✚ Some coccidiostats, such as ionophores and benzimidazoles, affect the calcium channels or the energy metabolism of the protozoan cells, causing them to lose their motility or viability.
- ✚ This mechanism is effective against most species of *Eimeria*.
- **Indications of coccidiostats:**
 - ✓ Used for various types of poultry production systems, such as broilers, layers, and breeders.
 - ✓ The dosage and duration of treatment depend on the type and severity of the infection and the poultry species.
 - ✓ Coccidiostats are usually given in feed or water as a preventive measure or a therapeutic intervention.
- **Side effects of coccidiostats:**
 - ✓ Reduced feed intake, reduced egg production, reduced egg quality, or drug residues in meat or eggs.
 - ✓ Coccidiostats should not be used in poultry that are pregnant, lactating (producing eggs), or hypersensitive to the drug.
- **Coccidiostats drugs**
 - ✓ **Amprolium:** Organic compound sold as a coccidiostat used in poultry
 - ✚ **MoA:** Competitive inhibition of thiamine uptake: prevents coccidia from utilizing thiamine. Pk: Rapidly eliminated (within few hours) from the body via kidneys.
 - ✚ **Indication:** Coccidiostat (for poultry).
 - » **In Poultry:** Good activity against and used as a therapeutic agent *E. tenella*, *E. acervulina* in poultry. It has weak activity against *E. maxima*, *E. mivati*, *E. necatrix*, or *E. brunetti*.
 - » **In cattle:** Approved for the treatment and prevention of *E. bovis* and *E. zurnii* in cattle and calves

✚ **Toxicity:** Wide margin of safety.

- **Ionophores**

- ✓ E.g monensin, salinomycin, lasalocid ...
- ✓ principal choice to control coccidiosis
- ✓ **MoA** = Disrupt ion gradients across the cell membrane of the parasite
- ✓ Monensin

✚ Used extensively in the beef and dairy industries to prevent **coccidiosis**, increase the production of propionic acid and prevent bloat.

- **Sulfonamides:** Inhibition of the folic acid pathway. Sulfonamides and trimetoprim are primarily used as antibiotics, but they are effective against some protozoal species as well.

B. Babesicides: Commonly used babesicides are diminazene aceturate , imidocarb dipropionate

- **Diminazene aceturate** (4,4'-(diazamino) dibenzamidine diacetate)

- ✓ Used in tropical countries for the treatment of animal trypanosomiasis and babesiosis.
Less susceptible to relapse

- **Imidocarb dipropionateis:** A urea derivative used in veterinary medicine as an antiprotozoal agent for the treatment and prophylaxis of infection with Babesia and other parasite.

C. Trypanocidals: A trypanocidal agent is an antiprotozoal agent that acts upon trypanosome parasites. E. g. Diminazene aceturate, ethidium, Isometamidium chloride

- **Diminazene aceturate**

- ✓ Most commonly used trypanocidal drug in **cattle, sheep and goats**, due to its activity against *T. congolense*, *T. vivax* and *T. brucei*.
- ✓ only applied as a curative agent and is not used for prophylaxis, as it is rapidly metabolized and excreted
- ✓ Diminazene residues may persist for several weeks in the edible tissues of cattle and other food-producing animals.
- ✓ For this reason it is advised that cattle and sheep destined for human consumption are subject to a 21–35 days pre-slaughter withdrawal while a 3-day milk discard period is recommended.

- **Ethidium bromide:**

- ✓ It is widely used in Africa to treat *T. congolense* and *T. vivax* infections in cattle, sheep and goats, in spite of its proven mutagenic and possible carcinogenic properties as a DNA intercalator.
- ✓ Administered at the dose of 1 mg kg⁻¹ by a single, deep intramuscular injection
- ✓ It blocks both kinetoplast and nuclear DNA replication in glossina by distorting and changing the double helix topology.
- **Isometamidium chloride:**
 - ✓ Synthesized by coupling ethidium with the diazotized *p*-aminobenzamide moiety of diminazene, modified with the amidine group in the *Meta* position.
 - ✓ It has both curative and prophylactic properties.
 - ✓ Isometamidium is used primarily to treat and prevent *T. congolense*, *T. brucei* and *T. vivax* infections in livestock in Africa.
- **Suramin sodium:**
 - ✓ It is the oldest trypanocide still in use.
 - ✓ Suramin is also the standard treatment for equine trypanosomiasis (*T. brucei*) being more effective than others.
 - ✓ The current treatment for camels and horses is 10 mg kg⁻¹, administered intravenously.
 - ✓ Is effective against *T. gambiense*, *T. rhodesiense*, *T. evansi*, *T. brucei*, *T. equiperdum*; especially in camels, horses
 - ✓ Suramin curbs glycolytic ATP production in *T. brucei* by inhibiting glycerol-3-phosphate oxidase and NAD⁺-dependent glycerol-3-phosphate dehydrogenase.

III. Ectoparasitocides

- Can be grouped according to structure and mechanism of action into:
 - ✓ Organochlorines, organophosphates and carbamates,
 - ✓ Pyrethrins and pyrethroids,
 - ✓ Avermectins and milbemycins,
 - ✓ Formamidines, insect growth regulators

A. Organochlorines

- Withdrawn in many parts of the world due to concerns regarding environmental persistence.
 - ✓ DDT (dichlorodiphenyltrichloroethane), DDE (dichlorodiphenyldichloroethane), and DDD (dicofol, methoxychlor); chlordane, aldrin, dieldrin, heptachlor, endrin, and tozaphene.

B. Organophosphates (OP)

- Organophosphates are a class of chemicals that have many uses, such as insecticides, medications, and nerve agents.
- Extremely toxic in animals and humans, causing an inhibition of AChE and other cholinesterases
- Topically used include:
 - ✓ Diazinon, dichlorvos, famphur, fenthion, malathion, trichlorfon, stirofos, phosmet, and propetamphos.
 - ✓ generally active against fly larvae, flies, lice, ticks, and mites on domestic livestock
 - ✓ Diazinon and propetamphos dip formulations for the control of psoroptic mange in sheep
- **MoA:** They work by interfering with the nervous system of insects and mammals, causing various symptoms and complications depending on the dose and exposure.
- **Indications:**
 - ✓ Are neurotoxic to parasites; some cause neurologic side effects in the host
 - ✓ Both endoparasitic and ectoparasitic activity
 - ✓ Narrow range of safety; not for use in heartworm-positive dogs
 - ✓ Effective against bots and a variety of nematodes

C. Organophosphate poisoning

- Can occur when a person is exposed to them for too long or at high levels.
- Can cause long-term health problems, such as metabolic disorders, cancer, neurological issues, fertility problems, and paralysis

- A global public health concern, especially for agricultural workers in developing countries
- Can be treated with antidotes, such as atropine and pralidoxime, which block the effects of the chemicals on the nervous system.

D. Pyrethrins and Synthetic Pyrethroids

- **MoA:** Interference with sodium channels of the parasite nerve axons, resulting in delayed repolarization, incoordination and convulsion and eventual paralysis
- **Activity:** Active against biting and nuisance flies, lice, and ticks on a domestic livestock
- **Common pyrethroids used include:** Bioallethrin, cypermethrin, deltamethrin, fenvalerate, flumethrin, lambda-cyhalothrin, phenothrin, and permethrin

✓ **Deltamethrin:** Synthetic pyrethroid

✚ **MoA:** Deltamethrin works by interacting with the octopamine receptors and the monoamine oxidase enzyme in the nervous system of the parasites, causing them to become overstimulated and paralyzed.

✚ **Indications:**

- » Used to control various types of external parasites in animals, such as mites, ticks, fleas, lice, and flies.
- » It is effective against many species of mites and ticks, such as Demodex, Sarcoptes, Otodectes, Rhipicephalus, Dermacentor, Ixodes, and Amblyomma. Deltamethrin can also control some species of flies, such as Musca, Stomoxys, Haematobia, and Lucilia.

✚ **Pk:** Can be applied as a dip, a spray, a pour-on, or an impregnated collar, depending on the type of parasite and the animal species. .

✚ **SE:**

- » Sedation, hypothermia (low body temperature), bradycardia (slow heart rate), hypotension (low blood pressure), vomiting, diarrhea, or allergic reactions.

- » It should not be used in animals that are pregnant, lactating (producing milk), or hypersensitive to the drug.

E. Macrocyclic Lactones: Avermectins: Active against a wide range of nematodes and arthropods and, are often referred to as endectocides

F. Formamidines

- **Amitraz:** The only formamidine used as an ectoparasiticide.
- **MoA:**
 - ✓ Act by inhibition of the enzyme monoamine oxidase (MAO) that metabolizes amine neurotransmitters in ticks and mites and
 - ✓ As an agonist at octopamine receptors octopamine is thought to modify tonic contractions in parasite muscles.
 - ✓ Amitraz has a relatively wide safety margin in mammals
- **PK:**
 - ✓ The dosage and duration of amitraz treatment depend on the type and severity of the infestation and the animal species.
 - ✓ Amitraz is available as a dip for control of demodicosis and scabies in dogs.
 - ✓ It is also available as an impregnated collar for control of ticks in dogs.
- **Indications:**
 - ✓ Used to treat infections caused by parasitic mites and ticks in animals
 - ✓ Demodicosis: an infection caused by the mite *Demodex canis*
 - ✓ Scabies: an infection caused by the mite *Sarcoptes scabiei*
 - ✓ Otodectic mange: an infection caused by the mite *Otodectes cynotis*
 - ✓ Tick infestation: an infestation caused by various species of ticks, such as *Rhipicephalus*, *Dermacentor*, *Ixodes*, and *Amblyomma*
- **SE:**
 - ✓ Sedation, hypothermia (low body temperature), bradycardia (slow heart rate), hypotension (low blood pressure), vomiting, diarrhea, or allergic reactions.
 - ✓ It should not be used in animals that are pregnant, lactating (producing milk), or hypersensitive to the drug.

- ✓ It should also be used with caution in animals that have diabetes, heart problems, or respiratory problems

IV. Mechanism action of antifungal drugs

- Fungi are eukaryotic, heterotrophic organisms that live as saprobes or parasites. They have:
 - ✓ Nucleus with well-defined nuclear membrane.
 - ✓ Rigid cell wall composed of glucan, chitin, glycoproteins, melanin which is different from peptidoglycan layer of bacterial cell wall.
 - ✓ **Antifungal Drugs classes**

I. Polyene antifungals

- Amphotericin B, nystatin, natamycin, candicidin
- **MoA:** Interact with ergosterol to form channels through the membrane, causing the cells to become leaky
 - ✓ Amphotericin B (*deoxycholate*)
 - » Fungicidal
 - » Remains the standard treatment for many severe, invasive fungal infections
 - » Indication: Active against mucormycosis, aspergillosis, blastomycosis, candidiasis, coccidioidomycosis, cryptococcosis
 - » SE: Nephrotoxicity, hypokalemia, thrombophlebitis.

II. Azoles

- Fungistatic
- Classified in to imidazoles (Commonly used imidazoles are clotrimazole, miconazole, and ketoconazole), and Triazoles (Itraconazole, fluconazole, voriconazole and Posaconazole)
 - ✓ **Ketoconazole:**
 - ✚ First orally active azole for systemic mycoses
 - ✚ Active against histoplasmosis, blastomycosis, candidiasis, coccidioides, dermatophytes
 - ✚ Largely replaced by Itraconazole.
 - ✚ Adverse effects: GI, steroid synthesis inhibitor, transient increase in liver enzymes.
 - ✓ **Itraconazole:**

- ✚ Well absorbed orally
- ✚ Active against blastomycosis, histoplasmosis (including AIDS associated ones), candidiasis, aspergilosis
- ✚ Adverse effects: GI, rash, hypokalemia, hypertension, edema, headache.
- ✓ **Fluconazole:**
 - ✚ Drug of choice for cryptococcus neoformans, candidiasis, coccidioidomycosis.
 - ✚ Adverse effects: GI, rash, hepatotoxicity.

III. Flucytosine

- Echinocandins: Caspofungin, Micafungin, Anidulafungin
- Terbinafine, Ciclopirox, Tolnaftate, Benzoic Acid, Copper sulphate, Zinc Sulfate

IV. Terbinafine

- Drug of choice for treating dermatophytes
- Better tolerated , requires shorter duration of therapy
- fungicidal
- Activity is limited to **C. albicans** and **dermatophytes**.
- Drug accumulates in skin, nails and fat.
- Adverse effects: GI, headache, rash

1.4. Pharmacokinetics and pharmacodynamics of veterinary drugs

1.4.1. Pharmacokinetics veterinary drugs

- It the quantitative study of drug movement in, through and out of the body.
- To produce its characteristic effects, a drug must be present in appropriate concentrations at its sites of action.
- Factors affecting Pharmacokinetics activists of drug includes
 - ✓ Age,
 - ✓ Gender,
 - ✓ Diet,
 - ✓ Environment,
 - ✓ Body weight and pregnancy,
 - ✓ Patient's pathophysiology,

- ✓ Genetics and
- ✓ Drug- drug or food-drug interactions.
- **Generally** It studies the processes of absorption, distribution, metabolism and excretion of drugs (how the body affects the drugs; movement or disposition of drugs in the body).

A. Absorption (oral or parenteral)

- It is the transfer of a drug from its site of administration to its site of action (or measurement).
- When a drug is applied to a body surface (e.g., G.I. tract, skin, etc.), its rate of absorption will determine the time for its maximal concentration in plasma and at the receptor to produce its peak effect.
- The factors that modify the absorption of drugs either related to the drug or to the patients:

I. Factors related with drugs

- Drug solubility
 - ✓ Dissolution is a requirements for Oral Absorption
 - ✓ Dissolution is the process by which a drug moves from the solid state into solution
 - ✓ The drug must be in solution to be absorbed
 - ✓ Insoluble drugs are not absorbed
- Degree of ionization
 - ✓ The greater the degree of ionization the lesser the absorption because the lesser Lipophilic properties of the drug ions.
 - ✚ Ionized drug---- polar---- less lipid soluble ---Less absorption
 - ✚ Unionized drug --- non-polar --- more lipid soluble ---better absorption
- Effect of pH
 - ✓ Most drugs are either weak acids or weak bases
 - ✓ Weakly acidic drugs are easily absorbed in a low pH medium such as in the stomach. Whereas weakly basic drugs are not absorbed until they reach the higher pH medium in the small intestine
- Pharmaceutical form
 - ✓ Rate of permeation dependent upon molecular weight
 - ✓ Large molecules have problems diffusing across membrane

- ✓ Drugs in aqueous solution are more rapidly absorbed than those given in oily solutions or suspension; and
- ✓ The smaller the particle size of the powder the more efficient is their absorption

II. Factors related to the patients

• Routes of administration

- ✓ Excellent from pulmonary alveoli,
- ✓ Very good from sublingual mucosa,
- ✓ Best from parenteral sites
- ✓ For oral drug, it is absorbed first from GIT either through passive diffusion or active transport

• State of health of absorbing surface

- ✓ Contact time at the absorption surface. E.g., diarrhea

B. Distribution

- It is the process by which the drug reversibly leaves the bloodstream and enters to interstitial (ECF) and/or cells or tissues/its site of action.
- The blood, total body water, extracellular, lymphatic and cerebrospinal fluids are involved in drug movement throughout the body.
- Depending upon its chemical and physical properties, the drug may be bound to plasma proteins or dissolved in body fat, delaying its progress to its sites of action or excretory mechanism.
- **Kinetics of drug distribution to several tissues depends on the following:**
 - ✓ Dose and route of administration
 - ✓ Lipid solubility of the drug
 - ✓ Ionization at physiological pH (dependent on pKa)
 - ✓ Extent of binding to plasma & tissue proteins
 - ✓ Blood flow rates through tissues and organs, e.g. thiopental sodium is redistributed to poorly perfused tissues (fat and muscles) after attaining high concentration in well perfused tissues (brain, liver and kidneys).

C. Metabolism

- Drugs undergo metabolic changes in the body that are directed primarily toward formation of metabolites that have physicochemical properties favourable for their excretion.
- Products of biotransformation are generally less lipid soluble and more polar in nature. These properties help excretion of drugs.
- Mostly hydrophilic drugs are not biotransformed. Example – Streptomycin, Neostigmine etc.
- Drug metabolism has been generally divided into two types of reactions, termed phase I and phase II reactions.

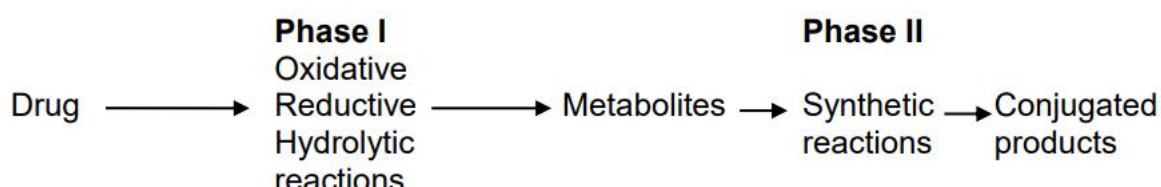


Figure 1.5: Phase of drug metabolism in liver

- The initial phase consists of reactions that can be classified as oxidative, reductive, and hydrolytic while the second phase includes the synthetic reactions (conjugations).]
- Phase I biotransformation usually unmask or introduce into the drug molecule polar groups such as – OH, –SH, –COOH, and –NH₂. These functional groups enable the compound to undergo conjugation with endogenous substances such as glucuronic acid, acetate (acetylation), sulphate (sulphuric acid ester formation) and various amino acids (primarily glutathione, cysteine and glycine). The drug conjugates formed are water soluble and almost invariably inactive pharmacologically.
- Metabolites of phase I biotransformation reactions may be pharmacologically active or inactive, but phase II metabolites are mostly inactive.
- **Biotransformation reactions may lead to the following:**
 - ✓ **Inactivation:** Most drugs and their active metabolites are rendered inactive or less active. For example, phenobarbitone, morphine, chloramphenicol etc.
 - ✓ **Active metabolite from an active drug:** Many drugs have been found to be partially converted to one or more active metabolite, the effects observed are the sum total of that due to the parent drug and its active metabolite (s).

- ✓ **Activation of the inactive drug:** Few drugs are inactive as such and need conversion in the body to one or more active metabolites. Such a drug is called a ‘Prodrug’. The pro drug may offer advantages over the active form in being more stable, having better bioavailability, or other desirable pharmacokinetic properties or less side effects and toxicity.

Examples: Prodrug	Active form
Levodopa	Dopamine
Prontosil	Sulfanilamide
Hetacillin	Ampicillin

D. Excretion

- The kidney is the most important organ for drug excretion but the liver, lung and skin are also involved in drug elimination.
- Drugs excreted in feces are mostly derived from unabsorbed, orally ingested drugs or from metabolites excreted in the bile and not reabsorbed by the intestine.
- The physical and chemical properties, especially the degree of ionization of the drug, are important in the rate of excretion.
- Excretion can be :

- ✓ **Renal Excretion:**

- ✚ In general, compounds that have low lipid solubility and those that are predominantly ionized in blood plasma are rapidly excreted by the kidneys.
- ✚ Carrier mediated excretion of drugs: Probenecid decreases the rate of elimination of Penicillin G and ceftiofur by reducing tubular secretion of the antibiotic.
- ✚ pH of the urine is also an important determinant of drug excretion. Acidic drugs are excreted more in alkaline urine whereas alkaline drugs are eliminated fast in acidic urine.
- ✚ Smaller molecular weight and ionized drugs are excreted unchanged in urine. Examples – Penicillin, cephalosporins, aminoglycosides, oxytetracyclines and most diuretics etc.

✓ **Biliary Excretion:**

- ✚ Apart from kidneys, liver also eliminates few drugs via bile.
 - ✚ High molecular weight compounds and presence of polar groups are the properties of compounds excreted by the liver via bile. Examples – Nafcillin, erythromycin, digitoxin, certain indigenous compounds like steroid hormones and conjugates of chloramphenicol, morphine and bilirubin are excreted via bile.
 - ✚ Enterohepatic circulation: Delays elimination of drugs. e.g. Tetracycline etc.
- ✓ **Excretion into milk:** Cow's milk is slightly acidic (pH 6.5 – 6.9) relative to plasma (pH 7.2 – 7.4) and therefore tends to concentrate basic fat soluble drugs.
- ✓ **Excretion from lungs:** Volatile anesthetics.
- ✓ **Saliva and sweat:** These routes of excretion are of minor importance. e.g. Potassium iodide, lithium iodide etc.

1.4.2. Pharmacodynamics

- The study of physiological and biochemical effects of drugs and how these effects relate to a drug's mechanism of action.
- Characterizes what a drug does to a patient.
- In contrast, the study of pharmacokinetics addresses what the patient's body does to a drug.

I. Factors Modifying Drug Action:

✓ **Species variation:**

- ✚ The response of a drug is not uniform among different species.
- ✚ Example: Morphine produces CNS depression in man and dog but excitation in cat.
- ✚ Rabbits can thrive on Belladonna leaves which are too toxic to other species.

✓ **Individual variation:**

- ✚ A drug at equal doses in similar animals will not produce identical magnitude of the response.
- ✚ This is due to normal biological variation between the individuals in a population, as determined by genetic factors.

- ✚ This is known as Idiosyncrasy. There may be some specific genetic defects which lead to discontinuous variation in drug responses.
- ✓ **Age of the animal:**
 - ✚ Very young (new born) and very old animals are more sensitive to the drug effects than the normal adults. In the former, the organs of biotransformation (liver) and excretion (kidneys) are not functionally developed to full capacity.
 - ✚ These mechanisms are defective in very old animals. The renal clearance of drugs in these animals is also poor (low GFR i.e. glomerular filtration rate).
 - ✚ Young and old animals need relatively low doses of a drug as compared to the adults.
- ✓ **Sex of the animal:**
 - ✚ Generally female animals are more sensitive to the effects of drugs than the males. This is also true for effects of toxic substances.
 - ✚ The relative resistance of males is due to presence of high concentrations of testosterone.
 - ✚ Gynaecomastia is a side effect (of ketoconazole, metoclopramide, chlorpromazine, digitalis etc.) that can occur only in men.
 - ✚ Ketoconazole causes loss of libido in men, not in women
- ✓ **Body weight of the animal:**
 - ✚ It indicates the extent of tissues that are exposed to the action of drugs. Therefore, the dose of a drug is based on the body weight of the animal.
 - ✚ While computing dose rates of drugs in ruminants, the weight of rumen contents should be taken in consideration.
 - ✚ The other conditions such as pregnancy (weight of foetus), oedema (fluid) and fat deposits (obese pigs) also need to be considered.
- ✓ **Route of administration:**
 - ✚ It governs the speed and intensity of drug response.
 - ✚ The onset of action following administration of drugs by different routes is in the order – I.V. > I.M. > I.P. > S.C. > I.D. > oral.

- ✚ A drug may have entirely different uses through different routes. For instance, MgSO₄ given orally causes purgation, applied on inflamed areas decreases swelling, while intravenously it may produce CNS depression and hypotension.
- ✓ **Time of administration:**
 - ✚ Subjective effect of a drug may be markedly affected by the setup in which it is taken.
 - ✚ Hypnotics taken at night and in quiet, familiar surroundings may work more easily.
 - ✚ It has been shown that corticosteroids taken as a single morning dose cause less pituitary-adrenal suppression
- ✓ **Psychological factor:**
 - ✚ Efficacy of a drug can be affected by patients belief's, attitudes and expectations
- ✓ **Drug Interactions:**
 - ✚ Drugs may modify the response of each other by pharmacokinetic or pharmacodynamic interaction between them.
 - ✚ Pharmacokinetic interactions: It refers to interactions between the drugs during absorption, distribution, metabolism and excretion processes, resulting in either increase or decrease in the concentration of the drug at the site of action
 - ✚ Pharmacodynamic interactions: These interactions occur due to the action of drugs at a common receptor site or at different sites, resulting in increase or decrease in response of a drug.
- ✓ **General state of health of animal:**
 - ✚ Generally weak and debilitated animals are more sensitive to drug effects than the normal healthy animals.
 - ✚ The reason being low hepatic glycogen stores in the former, contributing to reduced metabolic capacity of the organ.
 - ✚ Drugs having high fat affinity may cause toxic effects in lean animals. Similarly, fatty and obese animals (pigs) require more doses of a fat soluble drug than that is required for a lean animal in producing identical therapeutic effect.
 - ✚ Hepatic and renal diseases increase the intensity and prolong the effects of drugs.
- ✓ **Drug tolerance:**

- ✚ It means requirement of higher dose of a drug to produce a given response.
- ✚ Tolerance is widely occurring adaptive biological phenomenon.
- ✚ Drug tolerance may be natural or acquired.
 - » **Natural drug tolerance:** The individual/ species are inherently less sensitive to the drug. Examples: Rabbits are tolerant to atropine (due to the presence of atropinase enzyme). Chicks are tolerant to strychnine (due to underdevelopment of spinal cord in chicken).
 - » **Acquired drug tolerance:** It occurs by repeated use of a drug in an individual who was initially responsive.
 - » Body is capable of developing tolerance to most drugs, but the phenomenon is very recognized in the case of CNS depressants (e.g. morphine, barbiturate, tranquilizers etc.)

II. Terms related with drug action

- **Affinity:** describes the tendency of a drug to combine with a particular kind of a receptor whereas
- **Efficacy:** (or intrinsic activity) of a drug refers to the maximal effect the drug can produce.
- **Potency:**
 - ✓ It refers to the dose of a drug that must be administered to produce a particular effect of given intensity.
 - ✓ It is influenced by the affinity of a drug. It varies inversely with dose. It is a relative rather than an absolute expression of drug activity.
 - ✓ Potency of a drug is not necessarily correlated with its efficacy or safety and the most potent drug within a series is not necessarily clinically superior.
 - ✓ Low potency is a disadvantage only if the effective dose is so large that it is too costly to produce or too cumbersome to administer.
- **Selectivity:**
 - ✓ It depends on the capacity of a drug to preferentially produce a particular effect.
 - ✓ The characteristic effect of the drug is produced at lower doses than those required to elicit other responses.
- **Specificity:**

- ✓ When all the effects produced by a drug are due to a single mechanism of action, the drug is said to be specific.
- ✓ A specific drug acts at only one type of receptor, but may produce multiple pharmacological effects because of location of receptors in various organs.
- ✓ Example: Atropine is a specific drug in that its varied effects can be attributed to its antimuscarinic action.
- **Agonist:**
 - ✓ A drug that possesses affinity for a particular receptor and causes a change in the receptor that result in an observable effect.
 - ✓ Agonists are further categorized as:
 - ✚ Full agonist: Produces a maximal response by occupying all or a fraction of receptors. (Affinity=1, Efficacy=1)
 - ✚ Partial agonist: Produces less than a maximal response even when the drug occupies all of the receptors. A partial agonist has less intrinsic activity than a full agonist. (Affinity=1, Efficacy= 0 to 1)
 - ✚ Inverse agonist: Activates a receptor to produce an effect in the opposite direction to that of the well-recognized agonist. (Affinity=1, Efficacy= -1 to 0)
- **Antagonist:**
 - ✓ An antagonist is a drug that blocks the response produced by an agonist.
 - ✓ Antagonists interact with the receptor or other components of the effector mechanism, but antagonists are devoid of intrinsic activity (Affinity=1, Efficacy=0).
 - ✚ Antagonism can be classified as:
 - » **Competitive Antagonism:**
 - It is completely reversible; an increase in the concentration of the agonist in the immediate vicinity of its site of action or biophase will overcome the effect of the antagonist.
 - Example: Atropine (Antimuscarinic agent) Diphenhydramine (H1 receptor blocker) Propranolol (β -adrenergic blocker) Spironolactone (Aldosterone antagonist)
 - » **Non-competitive antagonism:**

- A non-competitive antagonist conceptually removes the receptor or response potential from the system.
- Addition of more agonist to the bio-phase of inhibited receptor will not overcome the antagonism achieved by a non-competitive antagonist. Thus, the agonist has no influence upon the degree of antagonism or its reversibility.
- Example: Platelet inhibiting action of aspirin (The thromboxane synthase enzyme of platelets is irreversibly inhibited by aspirin, a process that is reversed only by production of new platelets).

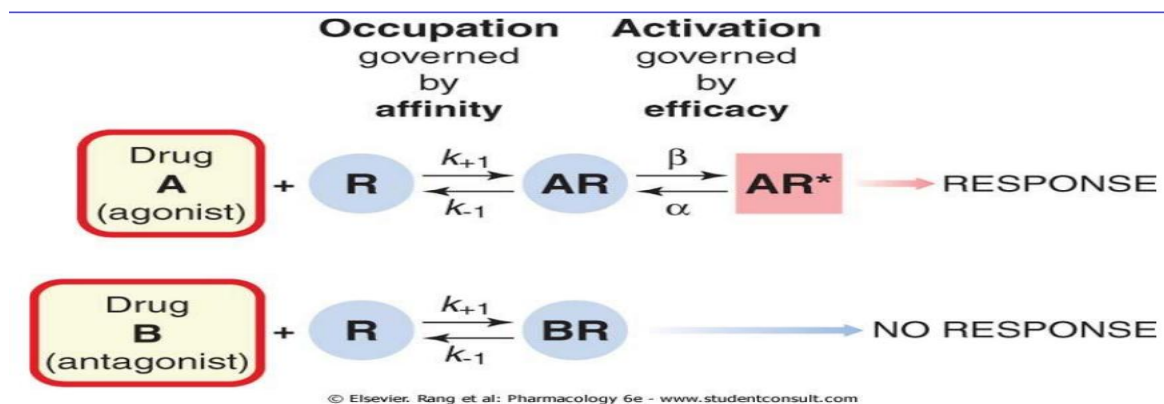


Figure: The distinction between drug binding and receptor activation. Ligand A is an agonist, because it leads to activation of the receptor (R), whereas ligand B is an antagonist

III. Combined Effect of Drugs

- When two or more drugs are given simultaneously or in quick succession, they may be either indifferent to each other or exhibit synergism or antagonism.
- The interaction may take place at pharmacokinetic level or at pharmacodynamic level.

✓ Synergism:

Derived from Greek words “Syn” = together, “ergon” = work

When the action of one drug is facilitated or increased by the other, they are said to be synergistic.

In a synergistic pair, both the drugs can have action in the same direction or given alone one may be inactive but still enhance the action of the other when given together.

✚ Synergism can be Additive or Supra-additive.

- » **Additive:** The effects of two drugs are in the same direction and simply add up i.e. Effect of drug A + B = Effect of drug A + Effect of drug B. E. g. Nitrous oxide + Ether = As general anaesthetic.
- » **Supra-additive (Potentiation):** The effect of the combination is greater than the individual effects of the components, i.e. Effect of drug A + B > Effect of drug A + Effect of drug B. E. g. Sulfonamide + Trimethoprim = Sequential blocking

✓ **Antagonism:**

✚ When one drug decreases or inhibits the action of another, they are said to be antagonistic, i.e. Effect of drugs A + B < Effect of drug A + Effect of drug B.

✚ Depending upon the mechanism involved, antagonism may be of following types –

- » **Physical antagonism:** Antagonism is based on physical property of drugs.
E. g. charcoal adsorbs alkaloids and can prevent their absorption. This phenomenon is employed in alkaloid poisonings.
- » **Chemical antagonism:** The two drugs react chemically and form an inactive product.
E. g. KMnO₄ oxidizes alkaloids (used for gastric lavage of drugs)
Chelating agents (like BAL, CaNa₂EDTA) complex with metals (like As, Pb etc.).
Drugs may react when mixed in the same syringe or infusion bottle. Example – Thiopentone Na + Succinylcholine chloride
- » **Physiological/ Functional antagonism:** Two drugs act on different receptors, or by different mechanisms, have opposite overt effects on the same physiological function i.e. have pharmacological effects in opposite direction.
E.g. Histamine and adrenaline on bronchial muscles and blood pressure, Glucagon and Insulin on blood sugar level.
- » **Antagonism by Receptor Block:** The antagonist interferes with binding of the agonist with its receptor or inhibits the generation of response consequent to such binding.

Examples

– ACh + Atropine, Morphine + Naloxone,

- Irreversible enzyme inhibitors that act similarly like aspirin, omeprazole, monoamine oxidase inhibitors
- Drugs such as verapamil and nifedipine prevent the influx of Ca^{2+} through the cell membrane and thus block non-specifically the contraction of smooth muscle produced by other drugs.

Self-check 1	Written test
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Name..... ID..... Date.....

Directions: Answer all the questions listed below.

Test I: much A to B (4 point)

Part I: Write “True” if the statement is correct or “False” if the statement is incorrect (3Points)

1. Over-use of injections when oral formulations would be more appropriate.

2. Dose is the amount of drug taken at any one time. It can be expressed as the weight of drug.
3. Dose can be given commonly in mg/kg body weight.

Part II: Match column “A” with appropriate answer from column “B” (9 Points)

<u>A</u>	<u>B</u>
_____ 1. Proprietary name	A. Official name
_____ 2. Empirical therapy	B. Trade name
_____ 3. Irrational drug use	C. Scientific name of drug
_____ 4. Generic name	D. What the drug dose to the patient
_____ 5. Pharmacokinetics	E. What the patient dose to the drug
_____ 6. Pharmacodynamics	F. Preventing the occurrence of a disease
_____ 7. Antimicrobials	G. Used against microorganism
_____ 8. Chemical name	H. Incorrect duration
_____ 9. Prophylactic or preventive therapy	I. Based on past experience

Part III. Choose the best answer for the following question (12 Points)

1. Which of the following is an intervention to reduce irrational drug use?
 - A. Use clinical guidelines
 - B. Supervision, audit and feedback
 - C. Independent information on drugs
 - D. Public education about drugs
 - E. All
2. The aims of using prescription papers are to achieve?
 - A. To maximize effectiveness
 - B. Minimize risks
 - C. Minimize costs
 - D. Respect the patient’s choices.
3. Which of the following drug is grouped under Penicillin antibiotics?
 - A. Methicillin
 - B. Oxacillin
 - C. Dicloxacillin
 - D. Cloxacillin
4. Which one is total synthesis Tetracycline Is?
 - A. Tigecycline
 - B. Doxycycline
 - C. Chlortetracycline
 - D. Demeclocyclin
5. Which of the following drug is not grouped under **Macrolides**?
 - A. Azithromycin,
 - B. Clarithromycin,
 - C. Erythromycin
 - D. None
6. Mechanism action of Sulfonamides is?
 - A. Inhibit cell wall of bacteria
 - B. Inhibit protein synthesis
 - C. Damage the cell membrane
 - D. Inhibit folic acid synthesis

Part IV: Short Answer Questions (16 points)

1. List the Characteristics irrational drug use (2 Points)

2. List Causes of irrational use of medicines (2 Points)
3. Define the following words? (7 Points)
 - A. Pharmacology
 - B. Drug
 - C. Antibiotics
 - D. Chemotherapy
 - E. Toxicology
 - F. Rational veterinary drug use
4. If the labeled dose of Oxytetracycline (10%) is 1ml/10kg, how much Oxytetracycline do you need to treat a sheep having 30kg body weight? (2 Points)
5. If the dose of a certain drug /Acaricide/ for hand spray is 100ml/100liter of H₂O and for the treatment of 5 goats if you assume 10 liter of water is sufficient. How much acaricide do you need to make the appropriate drug water solution (emulsion)? (2 Points)
6. What volume of formaldehyde is needed to prepare 100ml of 10% formalin from 40% formaldehyde solution? (2 Points)

Notice: Scoring rates above **20** is satisfactory and less than that is not satisfactory

LG #2

LO #2- Apply therapeutics drugs based on causative agent and body system

Instruction sheet 2

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- Use of veterinary drugs and control of drug hazards
- Administration of preventive and treatment drugs and fluids
- Implementation of OHS procedures as of Ethiopian VDFACA
- Classification of drugs acting on the different body system
- Veterinary drugs and chemicals for various diseases
- Environmental health policies and legislations to wastes dispose

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Recognize and report hazards in handling and use of veterinary drugs to designated personnel according to organizational policies and procedures.
- Implement OHS procedures in accordance with the Ethiopian Drug and Feed Administration and Control Authority.
- Identify and list drugs acting on the different body systems.
- Mention and explain category, mechanism of action, dosage form, dose, route of administration, therapeutic indication, side effect, residual effect and contraindication of commonly available veterinary drugs and chemicals for various disease causing agents.
- Prepare and implement prescription based on the rational diagnosis of the disease.
- Dispose wastes in line with environmental health policies and legislations.

Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

”

Information Sheet 2

2.1. Use of Veterinary Drugs and Control of Drug Hazards

2.1.1. Uses of Veterinary Drugs

- Veterinary drugs are used to prevent, diagnose, and treat diseases in animals. They play a crucial role in maintaining animal health and welfare. Here are some examples of veterinary drugs and their uses:
 - ✓ Amoxicillin: An antibacterial drug used to treat various infections.
 - ✓ Acarida for deworming of external parasites
 - ✓ Anthelmintic drug used to treat worm infestations.
 - ✓ Antiprotozoal drug: for treatment of protozoa
 - ✓ Antifungal drugs: for treatment of fungal disease
 - ✓ Antiviral drug: For treatment of viral infections
 - ✓ Used as a sedative, tranquilizer, and antiemetic (Acepromazine)
 - ✓ Antidepressant used to treat separation anxiety and excessive grooming in dogs and cats.
 - ✓ NSAID used to relieve pain and inflammation in dogs and cats.
 - ✓ An anti-inflammatory steroid (Dexamethasone).
 - ✓ Prophylactic agents: Sub-therapeutic doses of antibiotics via feed or drinking water for prevention of disease
 - ✓ Control of reproduction: Prostaglandins regulate fertility and breeding programs.
 - ✓ Preservation and processing of food: Food additives (antioxidants, coloring agents, stabilizers, sweeteners, tenderizers, etc.)
 - ✓ Growth promotion: antimicrobial agents administered at low or sub therapeutic dose and destroys or inhibits growth of microbes as infectious agents reduce the yields of food animal.

2.1.2. Control of Drug hazards

- It's important to assess the hazards specific to your workplace and develop hazard control strategies to mitigate those hazards. These may include the use of personal protective equipment (PPE), such as masks and respirators, when appropriate.

- Some of the hazards associated with drugs with their management are as follows:

A. Management of drug hypersensitivity or allergies primarily involves avoidance or discontinuation of the offending drug. Here are some strategies:

- **Avoidance or Discontinuation:** The most effective strategy is to avoid or discontinue the use of the causative drug. When available, alternative medications with unrelated chemical structures should be substituted.
- **Treatment of Symptoms:** Treatment is largely supportive and symptomatic. It may consist of topical corticosteroids and oral antihistamines for cutaneous symptoms such as hives and itching¹.
- **Anaphylaxis Treatment:** Anaphylaxis requires an immediate epinephrine injection. Hospital care also is necessary to maintain blood pressure and support breathing.
- **Desensitization:** In some cases, if it's medically necessary to take an allergy-causing drug, a process called desensitization might be used. This involves taking increasing amounts of the drug until the patient can tolerate the required dose.
- **Cross-Reactivity Consideration:** Cross-reactivity among drugs should be taken into consideration when choosing alternative agents.

B. Spillage of chemicals

- Managing spillage effectively is crucial to prevent harm to people and damage to the environment.
- Here are some key steps in spillage management:
 - ✓ **Prevention:** The best way to manage spills is to prevent them from happening. This can be achieved through careful handling of substances, regular equipment maintenance, and proper storage.
 - ✓ **Preparation:** Have a spill response plan in place. This should include procedures for preventing spills, stopping the source of a spill, containing and cleaning up a spill, disposing of contaminated materials, and training personnel.
 - ✓ **Containment:** If a spill occurs, the first step is to contain it to prevent it from spreading. This can be done using barriers or drainage areas.

- ✓ **Clean-up:** Cleaned up as quickly as possible using appropriate methods. For example, small spills of blood or other body fluids can be cleaned up by wiping the area with paper towels, then cleaning with warm water and detergent.
- ✓ **Disposal:** Contaminated materials used in the clean-up process should be disposed of properly to prevent further contamination.
- ✓ **Reporting:** All major chemical spill incidents should be reported through the appropriate channels.

C. Chemical burns can be serious and require immediate medical attention. Steps to manage a chemical burn:

- **Remove Dry Chemicals:** Put on gloves and brush off any remaining material.
- **Remove Contaminated Clothing or Jewelry:** Take off any clothing or jewelry that has come into contact with the chemical.
- **Rinse the Area:** Rinse the chemicals off for at least 20 minutes, in a shower if it's available. If it's an acid or chemical burn, carefully try to remove the chemical and any contaminated clothing, and rinse the affected area using as much clean water as possible.
- **Protect Your Eyes:** Be careful to avoid getting the chemical in your eyes while rinsing.
- **Bandage the Burn:** Cover the burn with a clean bandage¹.
- **Rinse Again if Needed:** If you feel more burning, rinse the area again for several more minutes

D. Management of hazards due to inhalation of drugs involves several strategies:

- **Proper Use of a Functioning Fume Hood:** This can protect laboratory workers from chemical exposure via inhalation
- **Use of Dust Masks and Respirators:** These should be used when a fume hood is not available.
- **Avoiding Bench Top Use of Hazardous Chemicals:** This can help to reduce the risk of inhalation.
- **Ensuring Chemical Containers Are Kept Tightly Capped:** This can prevent the release of hazardous vapors.

- **Prompt Clean-Up of Chemical Spills:** This can prevent the creation of hazardous aerosols.
- **Use of Engineering Controls:** Such as ventilated hoods and enclosures
- **Adoption of Administrative Controls:** Such as establishing safe handling policies, training, and routine training reviews for potentially exposed individuals.

2.2.Administration of Preventive and Treatment Drugs and Fluids

- The administration of preventive and treatment drugs and fluids in animals can vary depending on the specific medication, the animal species, and the condition being treated. Here are some general guidelines for administering drugs and fluids to animals:
 - ✓ **Consult with a Veterinarian:** Always consult with a veterinarian before administering any medication or fluids to your animal. They can provide specific instructions and dosage recommendations based on the animal's condition, weight, and individual needs.
 - ✓ **Oral Administration:** Many medications can be administered orally. This can involve giving tablets, capsules, or liquid formulations directly into the animal's mouth or mixing them with food. Follow the veterinarian's instructions regarding dosage and administration technique.
 - ✓ **Injectable Medications:** Some medications are administered via injections. This can include subcutaneous (under the skin), intramuscular (into the muscle), or intravenous (into the vein) injections. Injectable medications should be administered by trained individuals to ensure proper technique and minimize the risk of complications.
 - ✓ **Topical Administration:** Certain medications, such as flea and tick preventives or skin ointments, are applied topically to the animal's skin. Follow the instructions provided by the veterinarian or the product manufacturer for proper application.
 - ✓ **Intravenous Fluids:** In cases of dehydration, shock, or certain medical conditions, animals may require intravenous (IV) fluid therapy. IV fluids are administered through a sterile catheter inserted into a vein. This procedure is typically performed by a veterinarian or a trained veterinary professional.
 - ✓ **Follow Instructions:** It is essential to carefully follow the veterinarian's instructions regarding dosage, frequency, and duration of medication or fluid administration. Improper administration or dosage can have adverse effects on the animal's health.
 - ✓ **Proper Handling:** When administering medications or fluids, follow proper handling techniques to ensure safety for both you and the animal. This may involve wearing gloves, using sterile equipment, and maintaining a calm and controlled environment.

2.3.Implementation of OHS Procedures as of Ethiopian VDFACA

- The Occupational Health Safety procedures for the application of veterinary drugs involve several steps to ensure the safety of both the veterinary healthcare workers and the animals. Here are some key points:
- Some general information on the implementation of Occupational Health and Safety (OHS) procedures that are commonly followed in many countries. It's important to consult the Ethiopian Veterinary Drug, Food, and Control Authority (VDFACA) or other local regulatory bodies to obtain the most up-to-date and accurate information regarding OHS procedures in Ethiopia. Here are some general steps for implementing OHS procedures:
 - ✓ **Conduct a Risk Assessment:** Identify and assess potential hazards and risks in veterinary drug, food, and control facilities. This could include physical hazards, chemical exposures, biological agents, ergonomic issues, and other potential risks to employee health and safety.
 - ✓ **Develop OHS Policies and Procedures:** Establish clear policies and procedures that outline the responsibilities of employees, managers, and supervisors regarding occupational health and safety. This includes protocols for hazard identification, incident reporting, emergency response, and safety training.
 - ✓ **Provide Safety Training:** Ensure that all employees receive appropriate training on OHS procedures, including hazard identification, safe work practices, proper use of personal protective equipment (PPE), and emergency procedures. Training should be ongoing and refreshed periodically.
 - ✓ **Implement Safety Measures:** Take steps to eliminate or minimize hazards identified in the risk assessment. This may include implementing engineering controls (e.g., ventilation systems), providing appropriate PPE, establishing safe work practices, and ensuring the availability of first aid facilities.
 - ✓ **Establish Incident Reporting and Investigation Procedures:** Encourage employees to report incidents, accidents, near misses, and hazards promptly. Investigate incidents to identify root causes and implement corrective actions to prevent similar occurrences in the future.
 - ✓ **Regular Inspections and Audits:** Conduct regular inspections and audits to assess compliance with OHS procedures, identify areas for improvement, and ensure that safety measures are being followed consistently.
 - ✓ **Continuous Improvement:** Foster a culture of continuous improvement by reviewing and updating OHS procedures as needed, considering feedback from

employees, monitoring emerging hazards, and staying up-to-date with relevant regulations and best practices.

✓

2.4. Classification of Drugs Acting On the Different Body System

- Drugs can be classified based on the organ or system on which they act and their therapeutic, pharmacological, and chemical properties.

2.4.1. Drugs for the digestive system

- Drugs for the digestive system of animals can be classified into several categories based on their function:

I. Appetite Affecting Drugs:

- These drugs are used to stimulate or suppress appetite.
- Appetite disorders, particularly a lack of appetite, are very common in sick animals.
- There are several appetite-stimulating drugs; each drug works in a different way.
- **Appetite stimulant**
 - ✓ Used for Cachexic & anorexic animals, RF dogs
 - ✓ Drugs: Metoclopramide (promote gastric emptying), Diazepam-for cats (suppress part of the brain that controls satiety)

II. Emetics and Antiemetic's:

A. Emetic drugs

- Used to cause vomiting and are usually given in emergency situations after a pet has eaten a certain poison.
- They generally remove about 80% of the stomach contents.
- **Example of Emetics drugs**
 - ✓ **Apomorphine**
 - ✚ It is a morphine derivative that stimulates dopamine receptors in the CRTZ
 - ✚ Poorly absorbed after oral administration and is therefore usually administered topically in the conjunctival sac or parenterally.
 - ✚ Vomiting follows rapidly after intravenous administration, 5 to 10 minutes after intramuscular injection, and variably (10 to 20 minutes) after conjunctival administration.
 - ✚ **Clinical Uses**
 - » Used primarily for induction of vomiting in dogs.

- » It is considered by many to be the emetic of choice for dogs.
- » Its use in cats is controversial and possibly is contraindicated.
- ✓ **Apomorphine HCl** soluble tablets, from compounding pharmacies
 - ✚ **Adverse Side Effects:** These include protracted vomiting, restlessness, and depression.
- ✓ **Xylazine**
 - ✚ Although xylazine is not classified as an emetic, the label indicates that it induces vomiting in cats and occasionally in dogs.
 - ✚ Choice for inducing vomiting in cats.
- ✓ **Hydrogen peroxide**
 - ✚ Primary agent
 - ✚ A 3% solution of hydrogen peroxide can be used orally to induce vomiting.
 - ✚ Vomiting is induced by a direct irritant effect on the oropharynx and stomach lining.
 - ✚ Dogs and cats usually vomit within 5 to 10 minutes after administration of the hydrogen peroxide.
 - ✚ **Dosage:** Hydrogen peroxide 3% solution (1 teaspoon (5 mL) per 5 pounds of body weight not to exceed 45 mL, and this dose can be repeated once if not successful on the first attempt)
 - ✚ **Clinical Uses:** Hydrogen peroxide is used for the induction of vomiting in dogs, cats, pigs, and ferrets.
 - ✚ **Adverse Side Effects:** Aspiration pneumonia or gastric ulceration.
 - ✚ **Note:** Other emetics that have been used with various degrees of effectiveness include mustard and water, syrup of ipecac, and warm salt water.
 - ✚ **Contra-indications of emetics drug's**
 - » Acid or alkali poisoning: perforation & esophageal injury
 - » CNS stimulant poisoning: convulsions.
 - » Kerosene (Petroleum) poisoning: aspiration Pneumonia.
 - » Unconscious patients: aspiration of vomitus.
 - » Morphine or Phenothiazine poisoning: Ineffective.

»

B. Antiemetic drugs

- Used to stop vomiting. Continual vomiting is physically exhausting and can cause dehydration, acid-base and electrolyte disturbances, and aspiration pneumonia.
- The use of antiemetic is a form of symptomatic treatment because these drugs do not necessarily correct the underlying cause of the vomiting
- Antiemetic's usually are given parenteral because vomiting precludes use of the oral route
- **Note:**
 - ✓ Vomiting is caused by conditions that stimulate the emetic center of the brain.
 - ✓ It can be beneficial (in certain cases when a poison or toxin is eaten), but it can also be harmful in a sick or weakened animal that is likely to become dehydrated.
 - ✓ Not all animals have the ability to vomit, so drugs that induce vomiting should never be given unless directed by a veterinarian or animal poison control specialist.
 - ✓ Many cases of vomiting in small animals are self-limiting or can be controlled by withholding food and water for 24 to 48 hours.
- **Examples of Antiemetic drugs**
 - ✓ **Phenothiazine Derivatives**
 - ✚ Act centrally by blocking dopamine receptors in the CRTZ and possibly by direct inhibition of the vomiting center.
 - ✚ These agents are in widespread use.
 - ✚ They are very useful in preventing motion sickness in dogs and cats but may be less effective against irritant emetics
 - ✚ Common side effects include hypotension and sedation.
 - ✓ **Chlorpromazine:** A phenothiazine-derivative tranquilizer that has little popularity as a tranquilizer in veterinary medicine and is more often used as an antiemetic.
 - ✚ **Clinical Uses:** Chlorpromazine is used as an antiemetic in dogs and cats. It is more effective in dogs than in cats.
 - ✚ **Dosage Forms:** Tablets (various sizes), extended-release capsules (various sizes), oral solution (2 mg/mL, 30 mg/mL, and 100 mg/ML), Rectal

suppositories (25 mg and 100 mg), and injection (25 mg/mL in ampules and vials).

✚ **Adverse Side Effects:** These are primarily limited to sedation, ataxia, and hypotension.

✓ **Procainamide Derivatives:**

✚ Metoclopramide is a derivative of procainamide has antiemetic activities.

✚ Metoclopramide has a limited influence on GI secretions.

✚ This drug has a short half-life and may have to be administered often or in a continuous drip in severe cases of vomiting

✚ **Clinical Uses:** Used as an antiemetic for parvoviral enteritis, uremic vomiting, and vomiting associated with chemotherapy. It is also used to treat gastric motility disorder.

✚ **Dosage Forms:** Metoclopramide HCl (Reglan) tablets (5 and 10 mg), Oral solution (1 mg/ML) in containers of various sizes, and injection (5 mg/mL).

✚ **Adverse Side Effects:** The most common side effects in horses, dogs, and cats are behavioural or other disorders associated with the central nervous system (CNS). Constipation also may occur.

✓ **Antihistamines (H₁ Blockers)**

✚ Most effective as antiemetic's in dogs and cats when vomiting is a result of motion sickness or inner ear abnormalities. All antihistamines may cause sedation.

✚ **Dosage Forms**

» Trimethobenzamide HCl (Tigan): An antiemetic for use in dogs only.

» Dimenhydrinate (Dramamine). An antihistamine labeled for treatment of motion sickness in dogs and cats. It is available in tablet, liquid, and injectable forms.

» Diphenhydramine (Benadryl): Used in veterinary medicine as an antiemetic and for the treatment of motion sickness, pruritus, and allergic reactions. It is available in tablet, capsule, oral elixir, and injectable forms.

» Meclizine (Antivert): Used mainly in small animals for the treatment of motion sickness.

» Promethazine (Phenergan)

- ✓ **Serotonin Receptor (5-HT₃) Antagonists:** Blockade of these 5-HT₃ receptors causes antiemetic activity.

+ Dosage Forms

- » Ondansetron (Zofran): Used mainly as an antiemetic during chemotherapy and is noted for its special effectiveness during this application.
- » Dolasetron (Anzemet)
- » Granisetron (Kytril)

C. Antacids:

- These drugs neutralize stomach acid and are commonly used to treat ulcers.

Approach's to treatment

- **Reduction of gastric acid secretion**
 - ✓ Histamine (H₂ receptor) antagonist: Cimetidine, Ranitidine
 - ✓ Proton pump inhibitors: omeprazole, pantaprazole
 - ✓ Anticholinergics: pirenzepine, propantheline
 - ✓ Prostaglandin E₂ analogues: misoprostol
- **Neutralization of gastric acid (antacids)**
 - ✓ Systemic : Sodium bicarbonate, Sodium citrate
 - ✓ Non systemic : Magnesium hydroxide, Aluminium hydroxides
- **Ulcer potectives:**
 - ✓ Sucralfate- bind to and protects the ulcerated sites from acids, bile and pepsin activity
- **Anti-bacteria- H. pylori:** for humans
 - ✓ amoxicillin, clarithromycin, metronidazole, etc

D. Antidiarrheal Medications:

- Diarrhea is an increased frequency, volume or fluidity of the feces
- Treatment for diarrhea includes fluids, electrolyte (salt) replacement, maintenance of acid/base balance, and control of discomfort.

I. Mucosal Protectants and Adsorbents

- Drugs such as activated charcoal and bismuth subsalicylate are popular therapies for diarrhea.
- They work by protecting the lining of the intestines and/or absorbing the enterotoxins and endotoxins that cause some types of diarrhea.
- **Mucosal protectants**
 - ✓ Play a role in the management of peptic ulcer
 - ✓ Chelates or complexes, which coat the gastric mucosa preferentially at sites of ulceration by forming an adherent complex with proteins.
 - E.G. Sucralfate and tripotassium dicitratobismuthate
 - ✓ **Bismuth subsalicylate**- Act by stimulating the synthesis of protective mucus
 - ✚ Bismuth adsorbs bacterial enterotoxins and endotoxins and has a GI protective effect,
 - ✚ Salicylate component has anti-prostaglandin activity
- **Adsorbents**
 - ✓ Given orally, adsorb toxins from GIT, which may prevent irritation and erosion of the mucosa.
 - ✓ Bismuth salts, charcoal, zinc oxide and kaolin (aluminum silicate) are available for the treatment of non-specific diarrhea.
 - ✓ Kaolin does not adsorb *E.coli* enterotoxins and therefore has limited use in neonatal
 - ✓ Bismuth salts and activated charcoal may adsorb these toxins.

II. Non-steroidal Anti-inflammatory Drugs (NSAIDs)

- **NSAIDs** reduce the amount of prostaglandin produced in the body.
- Prostaglandin is a hormone involved in the process of muscle contraction.
- These drugs may be beneficial with some types of diarrhea.
- However, **NSAIDs** should be given cautiously because they can cause adverse gastrointestinal, liver, and kidney effects

III. Narcotic analgesics: opiates & opioids

E.g., Morphine & loperamide

- Increase segmental contractions
- Decrease intestinal secretions

- Increase intestinal absorption of luminal fluids, electrolytes & glucose

IV. Anti-parasitic drugs or dietary treatment: Example: metronidazole

V. Anti-inflammatories: Inflammatory Bowel Disease (IBD) in dogs & cats

- Prednisolone & methylprednisolone

VI. Immunosuppressant: IBD

- Cyclosporine, azathioprine, cyclophosphamide

VII. Laxatives & Cathartics:

- Laxatives & Cathartics increase the motility of the intestine or increase the bulk of feces. These drugs are administered
- To increase the passage of gastrointestinal contents,
- To cleanse the bowel before radiography or endoscopy,
- To eliminate toxins from the gastrointestinal tract, and
- To soften feces after intestinal or anal surgery.

Lubricant laxatives

- Act by coating the surface of the feces with a water- immiscible film and by increasing the water content of the feces to provide a lubricant action
- Soften and lubricate the fecal mass, which allows expulsion.

Example:

Liquid paraffin (LP) and white soft paraffin are commonly used

- LP is mixed with ginger to prevent inhalation, except which given by stomach tube.

Indication:

- Paraffin *Oral paste* 474mg/g for Constipation
- LP used in the treatment of Horse colic due to impaction
- ✓ **Contraindication (CI):** Prolonged use especially in young animals
- ✓ **Side effect:** Reduced absorption of nutrients; granulomatous lesions may develop with prolonged use

Bulk Laxatives

Non-absorbed synthetic or natural polysaccharide **cellulose derivatives**

- Imbibe water and increase the mass of non-digestible material in the intestine.

- ✓ Examples: methylcellulose, psyllium,, wheat bran,

Ispaghula takes up water in the GIT, thereby increasing the volume of the feces and promoting peristalsis.

- ✓ For management of chronic constipation
- ✓ used in the control of diarrhea-ability to increase fecal mass

Indications: Constipation

Osmotic Laxatives

- Are hypertonic solutions of poorly absorbed substances that retain and absorb water from the tissue into the intestinal lumen?
 - ✓ The resulting bowel distension promotes peristalsis.
- Magnesium sulfate (Epsom salt) is for both monogastric and ruminants
- Magnesium Hydroxide Mixture: Cream, 55 mg/ml
- Magnesium Sulfate (Epsom salt) Powder, 49 %, 90 %

Indications: Constipation

Precaution: Fluid should be available throughout treatment

CI: Abdominal pain, vomiting, intestinal obstruction

Warnings: Water must be available all times to avoid intestinal obstruction

Stimulant Cathartics:

- Stimulate intestinal motility via an irritant effect on the mucosa or stimulation of intramural nerve plexus.
- Also activate secretory mechanisms, provoking fluid accumulation in the GI lumen.
- Have potent effects, and excessive fluid and electrolyte loss can result.
 - ✓ **Vegetable oils-** hydrolyzed in the SI to irritating fatty acids
 - ✓ **Castor oil** is a potent cathartic hydrolyzed to release **ricinoleic acid**, which causes increased water secretion in the small intestine

Hyperosmotic Cathartics

- These drugs are poorly absorbed from the GIT and draw fluid into the intestine by osmosis.

- The fluid content of the feces increases, which causes intestinal distention and promotes peristalsis
- ✓ E.g. magnesium salts, sodium salts, and sugar alcohols

VI. Anthelmintic drugs

- Anthelmintic are drugs that combat parasitic worms, many of which infest the digestive tract.
- The most effective anthelmintic have a broad spectrum of activity against mature and immature parasites, are easy to give, inhibit reinfection for extended periods, have a wide margin of safety, and are compatible with other compounds.
- There are several classes of anthelmintics, including benzimidazoles and probenzimidazoles, salicylanilides and substituted phenols, imidazothiazoles, organophosphates, and macrocyclic lactones

2.4.2. Drugs acting on cardiovascular system

- Drugs acting on the cardiovascular system are a group of medications that are used to treat a variety of heart and blood vessel conditions. These drugs work by affecting the heart rate, blood pressure, and blood flow.
- There are many different types of drugs acting on the cardiovascular system, each with its own specific mechanism of action

A. Anti-Arrhythmic Drugs:

- Medications used to treat abnormal heart rhythms, also known as arrhythmias.
- Arrhythmias can be caused by a variety of factors, including heart disease, medications, and electrolyte imbalances.
- These drugs help to regulate the heartbeat. They work by blocking the electrical signals that control the heart's rhythm.

Example:

Quinidine Sulphate

- Primarily works by blocking the fast inward sodium current (INa).
- Quinidine's effect on INa is known as a 'use dependent block.

Uses:

- This medication is used to treat or prevent many types of irregular heartbeats (heart arrhythmias such as atrial fibrillation).
- Quinidine can greatly improve your ability to perform normal activities by decreasing the number of irregular heartbeats you have.

Procainamide Hydrochloride

Uses:

- It is mainly used for infiltration anesthesia, peripheral nerve block, and spinal block.
- Stability and storage conditions:
- Types of formulations: Injection and Tablet

Verapamil, Phenytoin Sodium

Uses:

- Verapamil is used to prevent chest pain caused by angina as well as irregular heart rhythms (arrhythmias).
 - It's also used to treat high blood pressure (hypertension).
 - If you have high blood pressure, taking verapamil helps to prevent: future heart disease.
- Stability and storage conditions:
- **Types of formulations:** Tablet

Amiodarone

Uses:

- Used to treat and prevent certain types of serious, life-threatening ventricular arrhythmias (a certain type of abnormal heart rhythm when other medications did not help or could not be tolerated.
- Stability and storage conditions:
- It should be kept in a cool, dry place and stored at room temperature. Direct sunlight is prohibited.
- Types of formulations:
- Tablet

Sotalol

Uses:

- It's used to treat atrial fibrillation and other conditions that cause an irregular heartbeat (arrhythmia).
 - This medication is used to treat a serious (possibly life-threatening) type of fast heartbeat called sustained ventricular tachycardia.
- Stability and storage conditions:

- It should be kept in a cool, dry place and stored at room temperature. Direct sunlight is prohibited.
- Types of formulations: Tablet

B. Anti-Hypertensive Agents:

- A class of drugs that are used to treat hypertension, also known as high blood pressure.
- Hypertension is a major risk factor for heart disease, stroke, kidney disease, and other health problems.
- These drugs lower blood pressure. They work by relaxing blood vessels, making it easier for blood to flow through them.

Example:

Propranolol

- It is a beta-adrenergic receptor antagonist

Uses: Used to treat a variety of conditions, including:

- High blood pressure (hypertension)
- Angina (chest pain)
- Arrhythmias (irregular heartbeats)
- Hyperthyroidism (overactive thyroid)
- Proliferating infantile hemangioma (a type of birthmark)

Types of Formulations: • Tablets • Capsules • Extended-release tablets • Oral solution • Injection

Captopril

Uses:

- Captopril is an angiotensin-converting enzyme (ACE) inhibitor.
- ACE inhibitors work by blocking the production of a hormone called angiotensin II.
- Angiotensin II is a powerful vasoconstrictor, which means that it causes blood vessels to narrow.
- By blocking the production of angiotensin II, captopril helps to relax blood vessels and lower blood pressure Ramipril

Other Anti-Hypertensive Agents includes:

- Methyldopate Hydrochloride
- Clonidine Hydrochloride
- Hydralazine Hydrochloride
- Nifedipine

C. Antianginal Agents:

- Class of drugs used to treat angina, a condition that causes chest pain due to a lack of oxygen to the heart.
- These are drugs used to treat angina pectoris, a chest pain caused by reduced blood flow to the heart.
-

Isosorbide Dinitrate

Uses:

- A prescription medication used to treat angina pectoris (chest pain).
- It works by relaxing the blood vessels, which increases blood flow to the heart.
- Types of formulations: • Tablets • Capsules • Sublingual

D. Cardiac Glycosides -Digoxin and Digitoxin

- Drugs which increase the efficiency of heart are called cardio tonic drugs

Pharmacological Actions

- Due to increase in force of contraction, cause a decrease in systolic time (duration of contraction).
- Increase in stroke volume
- Decrease in end diastolic volume
- Increase in automaticity of heart (decrease in transmembrane electrical potential and cardiac refractory period)
- All these effects lead to decrease in heart size.

Adverse effects/toxicity of cardiac glycosides

- Nausea and vomiting, Anorexia, Bradycardia, Cardiac arrhythmias, Headache, Blurred vision, Photophobia.....

2. 4. 3. Drugs acting on the blood and blood forming organs

A. Anticoagulants

Heparin

PK:

- Not absorbed after oral administration
- IV=immediate anticoagulant effect
- IM: should be avoided because of unpredictable absorption rates, local bleeding and irritation.

PK:

- Inhibits both in vitro and in vivo clotting factor.
- High dose can interfere with platelet aggregation and thereby prolong bleeding time.
- Protamine (Heparin Antagonist) = Specific heparin antagonist

Vitamin K antagonists

Example: Coumarin derivatives & indanediones

- They are orally active anticoagulants.
- They resemble vitamin K
- Importance in veterinary is due to their toxicity.
- Warfarin is the most reliable member of this group.

PK:

- Its sodium salt form
- Has 100% Bioavailability.
- Displacement of warfarin results in bleeding.

ADRs

- Hemorrhage
- Rare: diarrhoea, SI necrosis, urticaria, skin necrosis, purple toes, and dermatitis.

CIs

- GIT ulcer, thrombocytopenia, hepatic or renal diseases; recent surgery, bacterial endocarditis, pregnancy.
- Warfarin can cross the placenta and may cause teratogenicity and hemorrhage in the fetus.

Antithrombotic

Aspirin

- Inhibit platelet aggregation with low dose.

B. Hemostatics

Epinephrine and Norepinephrine

- Topical
- used as haemostatic's by means of their vaso-constrictive effects

Vitamin K

- Systemic hemostatics

Indication

- For the treatment of Rodenticides poisoning, Moldy clover poisoning (dicumarol), and Sulfaquinoxaline toxicity

C. Hematinic drugs (Antianemics)

- drugs or other agent that increases RBC production: Hgb synthesis & + bone marrow RBC formation

Vitamin B12 (Cyanocobalamin)

- Enteral absorption requires intrinsic factor from parietal cells, gastric acid, pepsin or pancreatic duct cells.
- The complex formed undergoes endocytosis
- Bound to *transcobalamin*, and transported for storage (liver) or uptake into tissues.
- Not degraded in the body; mainly excreted in bile and undergoes considerable enterohepatic circulation.
- Only trace amounts are normally lost in the urine and stool.

Iron Compounds

- Not all iron ingested is equally absorbable.
- **Fe³⁺** is virtually not taken up from the SI
- **Fe²⁺** is markedly better **absorbed**.

Factors facilitating iron absorption

- Acid; by favoring dissolution and reduction of Fe³⁺,

- Reducing substances: ascorbic acid, amino acid containing SH,
- Meat: probably by increasing HCl secretion.

Iron Preparations

Oral iron Preparations: Ferrous sulphate, Ferrous gluconate, Ferrous fumarate

- Absorption higher when taken in an empty stomach
- Oral adm is advantageous: impossible to overload the body with iron through an intact mucosa because of its demand-regulated absorption (mucosal block).

Parenteral iron preparations:

- Fe dextran, Fe-sorbitol-citric acid complex, Fe-sucrose complex, Fe-Na gluconate complex.

Parenteral preparations should be only when:

- Oral iron is not tolerated,
- There is malabsorption,
- Severe deficiency with chronic bleeding,
- Erythropoietin is used to induce rapid erythropoiesis.

2.4.4. Drugs that act on the respiratory system

A. Antitussive Drugs in Animals

Morphine

- An effective antitussive at doses lower than those that produce analgesia and sedation.
- It is not commonly used for antitussive activity because of adverse effects and the potential for abuse and addiction.
- Morphine has poor oral bioavailability because of a significant first-pass effect by the liver.

Codeine

- Codeine phosphate and codeine sulfate are found in many antitussive preparations, including tablets, liquids, and syrups.
- Codeine has analgesic effects that are about one-tenth that of morphine; however, its antitussive potency is approximately equal to that of morphine.
- The adverse effects of codeine are considerably less than those that occur with morphine at antitussive doses.

- Adverse reactions (especially in cats) are exhibited as excitement, muscular spasms, convulsions, respiratory depression, sedation, and constipation.
- Codeine should not be used after GI tract surgery because of its effects on intestinal motility.

Hydrocodone

- Chemically and pharmacologically similar to codeine but more potent.
- It is combined with an anticholinergic drug (homatropine) to discourage abuse by humans.
- It is prescribed for dogs with chronic cough due to canine bronchitis; however, information on its use is mainly anecdotal.

Butorphanol

- An opioid agonist-antagonist is used as an analgesic and antitussive in dogs.
- As an antitussive in dogs, butorphanol is 4 times more potent than morphine and 100 times more potent than codeine.
- At antitussive dosages, it may produce considerable sedation in dogs.
- Because butorphanol has poor bioavailability, the oral dose in dogs is 10 times the subcutaneous dose.
- In cats, butorphanol is primarily used as an injectable analgesic.
- In some cats, it may cause pain on injection, as well as mydriasis, disorientation, swallowing/licking, and sedation

Maropitant

- Has potential for use as a cough suppressant.
- Have antitussive effects in canine and feline models.
- In a study of dogs with chronic bronchitis, maropitant given at 2 mg/kg, PO, every 48 hours for 14 days reduced frequency of coughing.

Table 2.1: Summary of dosage for **Antitussive Drugs in Animals**

Drug	Dosage
Codeine	Dogs: 1–2 mg/kg, PO, q 6–12 h
Hydrocodone	Dogs: 0.25 mg/kg, PO, q 6–12 h
Butorphanol	Dogs: 0.055–0.11 mg/kg, SC, q 6–12 h, or 0.55–1.1 mg/kg, PO, q 6–12 h Cats: 0.1–0.4 mg/kg, SC, q 6–12 h
Maropitant	Dogs: 2 mg/kg, PO, q 48 h

B. Respiratory Stimulants in Animals

Doxapram

- Stimulates the medullary respiratory center and the chemoreceptors of the carotid artery and aorta to increase tidal volume.
- Other areas of the CNS are stimulated only when high doses are administered.
- It is used primarily in emergency situations during anesthesia or to decrease the respiratory depressant effects of opiates and barbiturates.
- Recommended dosages are 1–5 mg/kg, IV, in dogs and cats, or 1–2 drops under the tongue of apneic neonates. In adult horses, the dosage is 0.5–1 mg/kg, IV. Foals are dosed carefully at 0.02–0.05 mg/kg/h, IV.

C. Bronchodilators

- In certain airway disorders, breathing becomes difficult because the muscles surrounding the airways constrict.
- Bronchodilators relieve this constriction by relaxing the muscles, which opens the airways and allows air to move more easily in and out of the lungs.
- Bronchodilators may also help relieve inflammation and clear mucus from the lungs.

There are several categories of bronchodilators:

Beta-adrenergic agonists

- Have beneficial effects in the treatment of bronchoconstrictive airway diseases.
- Relax the bronchial smooth muscle, decrease the inflammatory response in the airways, and help the cilia clear mucus from the respiratory tract.
- They are used commonly in people for the treatment of asthma.

- In animals, they are used for airway disease, allergic bronchitis, and similar diseases such as feline “asthma,” recurrent airway obstruction (RAO) (“heaves”), and inflammatory airway disease (IAD) in horses

Example

Epinephrine (Adrenaline)

- Short-Acting Nonspecific Bronchodilators
- Considered the drug of choice for the emergency treatment of life-threatening bronchoconstriction, such as during an anaphylactic reaction
- The nonspecific stimulation of other receptors and its short duration make it unsuitable for long-term use.
- The dose used in animals is 10 µg/kg IM or IV administered as a one-time treatment, or repeated in 15 minutes
- Formulations are available in either 1: 10,000 concentrations (0.01%, 0.1 mg/ml) or 1: 1,000 concentrations (0.1%, 1 mg/ml). Duration of action is short (less than 1 hour).

Terbutaline:

- It is longer acting (6 to 8 hours).
- It may be injected subcutaneously to relieve an acute episode of bronchoconstriction.
- The dose in people is approximately 3.5 µg/kg, repeated once per hour
- The oral dose is 2.5 mg/dog q 8 h for dogs, and 0.625 mg (one-quarter of a 2.5-mg tablet) q 12 h for cats (e.g., cats with feline “asthma”)
- It is available as 2.5 and 5-mg tablets and 0.8 mg/ml (820 µ/ml) injection. The injection can be helpful for treatment of acute bronchoconstriction. The dose for injection in cats is 0.01 mg/kg IV or IM.
- Used in horses to treat recurrent airway obstruction (RAO), this is a bronchoconstrictive disease in horses analogous to asthma in people. It is not absorbed after oral administration in horses; therefore, injectable preparations must be used.

Methylxanthines (Xanthines)

- Decrease the inflammatory response in the airways and help the cilia clear mucus from the respiratory system.
- The methylxanthines are classified as CNS stimulants
- Relax bronchial smooth muscle and produce anti-inflammatory effects that are unrelated to the bronchodilating properties.

Anticholinergic drugs

- Effective bronchodilators that act by reducing the sensitivity of irritant receptors and inhibiting smooth muscles in the respiratory tract.
- Atropine and glycopyrrolate are effective bronchodilators.
- Anticholinergic drugs inhibit vagal-mediated cholinergic smooth muscle tone in the respiratory tract because asthmatic individuals have excessive stimulation of cholinergic receptors.

D. Expectorants and Mucus-producing Drugs

- Expectorants and mucus-producing drugs make coughs more productive.
- Increase respiratory secretions or sputum volume and reduce the viscosity of sputum.
- They do this by increasing the amount of bronchial secretions, in addition to thinning the secretions.
- This clears the airways and eases breathing.
- They are usually given by mouth, although some can be nebulized and inhaled.
- Used to increase the output of bronchial secretions, enhance the clearance of bronchial exudate, and promote a productive cough.
- Direct stimulants of respiratory secretions include the volatile oils, such as eucalyptus oil and oil of lemon. They are believed to directly increase respiratory tract secretions.

2. 4. 5. Urinary system drugs

Many different types of drugs are used in the management of renal disease and urinary system disorders.

- Some urinary drugs directly influence urine production and electrolyte balance.
- Others maintain blood pressure and reduce urinary system disease

A. Diuretics (Urine producing)

Diuretics increase the volume of urine excreted by the kidneys and thus promote the release of water from the tissues.

- This process called diuresis lowers the fluid volume in tissues.

- The two main purposes of diuretic use are to decrease edema and to lower blood pressure.
- Diuretics effectively reduce the edema associated with these conditions, as well as edemas of nonspecific nature, pulmonary edema, pulmonary congestion, and any pathological accumulation of non-inflammatory liquid.
- Urine formation is “Where sodium goes, water will follow.”
- The kidneys secrete and reabsorb sodium and chloride ions as they make urine.
- Diuretics block the reabsorption of these ions, so the sodium has nowhere to go but out of the kidneys and into the urinary bladder.
- Water then follows the sodium out of the kidneys and into the urinary bladder, and diuresis occurs.

The Classes of Diuretics

I. Thiazides

- Thiazides are diuretics that act directly on the distal convoluted tubules to block sodium reabsorption and promote chloride ion excretion.

Drugs:

- Hydrochlorothiazide, the standard thiazide drug, is given intravenously, intramuscularly, and orally to small animals, horses, and cattle.
- Other Veterinary thiazides include chlorothiazide, hydroflumethiazide , and bendroflumethiazide .

II. Loop agent

- Furosemide (Lasix),
- Others: bumetanide (Bumex), torsemide (Demadex), pretanide and ethacrynic acid (Edecrin).
- Furosemide and ethacrynic acid, two loop diuretics, are potent and effective drugs that block absorption of the following ions: chloride, potassium, calcium, hydrogen, magnesium, and bicarbonate.

Pharmacokinetics:

- Rapidly absorbed
- Diuretic response is extremely rapid following IV injection.

Pharmacodynamics

- Reduce the reabsorption of NaCl,
- Induce renal prostaglandin synthesis,
- NSAIDs can interfere with the actions of the loop diuretics by reducing prostaglandin synthesis in the kidney.

Clinical Indication

- Acute pulmonary edema,
- Other edematous conditions: RF, CHF, cirrhosis,
- Acute hypercalcemia ,
- Hyperkalemia,
- Acute renal failure, and
- Anion toxicity: ingestion of bromide, floride, iodide

ADR:

- Hypernatremia
- Hypokalemic Metabolic Alkalosis
- Ototoxicity (damage to the tight cell junctions in the blood vessels in the stria vascularis resulting in temporary disruption of the blood-cochlear barrier which increases the permeability of the lateral wall to ototoxic drugs)
- Hyperuricemia= UA reabsorption
- Hypomagnesemia and rarely hypocalcemia
- Allergic rxns, leukopnoea, pancreatitis, thrombocytopenia

CI:

- Furosemide, bumetanide, and torsemide may demonstrate cross-reactivity in patients who are sensitive to other sulfonamides.

SE:

- Electrolyte imbalances, especially hypokalemia, which can lead to cardiac arrhythmias.
- Parenteral administration produces diuresis immediately on a fully functional kidney

III. Osmotic agent

MoA:

- Osmotic diuretics increase the osmolality (concentration) of the filtrate in the renal tubules.
- This results in excretion of sodium, chloride, potassium, and water.

Indication:

- To Increase Urine Volume in ARF
- Reduction of Intracranial and Intraocular Pressure
 - Osmotic diuretics alter Starling forces so that water leaves cells and reduces intracellular volume.
- Management of poisoning= as irrigation fluid, mannitol
- Cerebral edema due to toxoplasmosis= decrease CSF

PK: Osmotic diuretics are **poorly absorbed**, they must be given parenteral.

- Mannitol causes **osmotic diarrhea** if administered orally,
 - ✓ Not metabolized
 - ✓ Excreted primarily by GF within 30–60 minutes, without any important tubular reabsorption or secretion.

PD:

- Are freely filtered at the glomerulus and increase tubular tonicity. Thus, they prevent passive water reabsorption
- They also *oppose* the action of ADH in the collecting tubule.
- Relatively large amount of water is lost than natriuretic eventually leading to hypernatremia.

SE:

- Dehydration and Hypernatremia, Headache, nausea, and vomiting

Table 1.2: Summary of diuretics drugs

Nephron Segment	Diuretic
Proximal convoluted tubule	CA inhibitors (e.g., acetazolamide) Osmotic agents (e.g., mannitol) Xanthines (e.g., aminophylline)
Ascending loop of Henle	Loop agents (e.g., furosemide) Osmotic agents
Early distal convoluted tubule	Thiazides (e.g., hydrochlorothiazide)
Late distal tubule and collecting duct	K ⁺ -sparing diuretics (e.g., triamterene or spironolactone)

2. 4. 6. Drugs for reproductive system of animals

Drugs used to regulate or control the reproductive system are often naturally occurring hormones or chemical modifications of hormones.

A. Gonadotropin-releasing hormone (GnRH) and its analogues

- Commonly used for control of ovarian follicular dynamics and **ovulation induction in cattle**, ovulation induction of mature follicles **in horses**, treatment of follicular cysts **in cattle** or **in dogs**, and
- Stimulation of libido or testicular function (eg, in testing for **cryptorchidism**) in horses, dogs, cats, and camelids.
- Implants of GnRH analogues (eg, deslorelin acetate) are effective for induction of estrus and ovulation in mares and bitches and, in higher doses over a prolonged period, are useful for contraception in male and female animals (such as dogs) by mediating longterm (≥ 12 months) reversible infertility caused by down regulation of GnRH receptors.

B. Follicle-stimulating hormone (FSH):

- Usually extracted from animal pituitary glands and stimulates follicular growth and estrogen production in females and spermatogenesis in males.
- Used for superovulation of several domestic species.
- Used for induction of a fertile estrus in anestrus bitches and queens.

- Prolonged FSH use or use of higher doses can cause adverse effects such as cystic endometrial hyperplasia and follicular cysts.

C. Equine chorionic gonadotropin (eCG)

- Has FSH activity in most species and is used to induce ovarian follicular growth and to induce estrus.
- Used in superovulation protocols in ruminants.
- Treatment to induce estrus in small animals experiencing anestrus.

D. Prostaglandins

PGF2alpha and its analogs

- Used mainly for their luteolytic effects to induce a predictable onset of estrus or to synchronize estrus in a variety of species.
- Used for termination of pregnancy, either alone (in horses or pigs) or in combination with corticosteroids (in cattle or sheep) or dopaminergic agents (in dogs).
- These compounds also cause marked uterine contractions, which may be useful for expulsion of uterine contents in pathologic conditions (eg, pyometra) in horses, pigs, ruminants, and small animals.
- The most common compounds used include dinoprost, a naturally occurring PGF2alpha, and cloprostenol, a synthetic prostaglandin analog.
- Common transient adverse effects in horses include sweating and abdominal discomfort (colic); increased salivation may occur in cattle.
- Small animals treated with prostaglandins may have increased salivation, vomiting, or diarrhea. The extent of the adverse effects may vary by patient and by the formulation used.

PGE2 (misoprostol)

- Applied topically to the cervical lumen or external cervical os in mares before induction of parturition or abortion or to assist in cervical dilation to facilitate the expulsion of intrauterine fluid.
- Applied topically to the serosal surface of the oviduct in mares to facilitate the passage of in vivo embryos by dilating the oviduct to resolve oviduct blockages.

- In dogs, used intra vaginally to assist with cervical relaxation

Oxytocin

- Used in a variety of species to promote milk letdown, as an adjunctive treatment of mastitis or agalactia, and to cause uterine contraction either to induce (or supplement) labor or to enhance uterine contraction for expulsion of uterine fluid or fetal membranes.
- It is administered parenteral (IV, IM, SC).
- In mares, oxytocin (60 IU, IM, every 24 hours for 30 days; or 60 IU, IM, every 24 hours, for 7–14 days when initiated after ovulation detection) has been administered to prevent estrus by prolonging the period of diestrus and CL function.

Melatonin

- May also be used to regulate reproduction in species that are seasonal breeders (small ruminants, cats, and horses).
- Species with breeding seasons respond to increases in melatonin by either increasing endogenous GnRH secretion (short-day breeders such as sheep or goats) or decreasing GnRH secretion (long-day breeders such as horses or cats).

2. 4. 6. Drugs act on Central nerve system of animals

Anesthetics

- Anesthetics are drugs that used to induce anesthesia in whole body (general anesthetics) or local area of the body (local anesthetics). It is divided into:

A. General Anesthetics

B. Local Anesthetics

A. General Anesthetics

- Drugs used to induce general anesthesia which is characterized by loss of sensation that achieved by reversible depression of the brain.

Clinical characteristics of general anesthesia:

- Loss of sensation.
- Unconsciousness.
- Muscle relaxation.
- Inhibition of reflexes.

Ideal anesthetics:

- Nonirritant and without bad odor.
- Potent with smooth induction and recovery.
- Wide margin of safety.
- Good muscle relaxation.
- Does not damage the CNS with minimal side effects.
- Inexpensive and non-explosive.
- Ease of administration.

A. General Anesthetics

- Inhalational Anesthetics
- Injectable Anesthetics

I. Inhalational Anesthetics

- Inhalational anesthetics are gases or vapors that diffuses readily across the pulmonary alveoli and then to the brain to produce anesthesia.
- The transfer of inhalational anesthetics depends on the partial pressure of anesthetics and its solubility in the blood.
- Lipid solubility is important for induction of anesthesia and recovery from it. High lipid solubility means long induction and long recovery periods of anesthesia (e.g. Ether) while low lipid solubility means rapid induction and fast recovery (e.g. Halothane).

Inhalational Anesthetics are:

- Gaseous like N₂O and Cyclopropane.
- Volatile like Halothane, Methoxyflurane, Enflurane and Ether

General mechanisms of action of inhalational anesthetics:

- Interaction with lipid molecules of the cell membrane.
- Interaction with proteins of the neuronal membrane.
- Interaction with water molecules of the cell membranes.

Halothane

- It is nonirritant, nonflammable, non-explosive and potent. It depresses all the functions of the brain. It bypasses stage II. It has rapid induction and fast recovery because it is low lipid soluble.

Side effects:

- Respiratory depression,
- Depresses cardiac function,
- Sensitizes the heart to catecholamines,
- Decreases calcium ion binding to cardiac muscle and causes bradycardia

Contraindication:

- Uses of epinephrine and norepinephrine are contraindicated in case of halothane anesthesia.

Advantages:

- It is potent and has rapid induction and rapid recovery from anesthesia.
- It produces good muscle relaxation

Disadvantages: 1. Cardio-pulmonary depression. 2. Expensive. 3. Contraindicated in cardiac diseases.

Nitrous oxide (N₂O)

- It is colorless, odorless, nonirritant and nonflammable. It is usually given as 80% N₂O and 20% O₂.
- It produces stage I and II anesthesia. It produces good analgesia, muscle relaxant is needed and O₂
- Given to prevent hypoxia. It is characterized by low lipid solubility so that, the induction and
- Recovery from anesthesia is fast. It is used with other anesthetic because it has low potency

Ether

- It is flammable, highly explosive and used in laboratory animals. It causes respiratory irritation and produces all stages of anesthesia.

Chloroform

- It is used in laboratory animals but causes liver toxicity so that, not used in other animal species.

II. Injectable Anesthetics

Advantages of Injectable anesthesia:

- Minimal apparatus used,

- Ease of induction,
- Useful in head surgery,
- Diminish the side effects of anesthetic in case of respiratory diseases, debility and shock.

Disadvantages of Injectable anesthesia:

- The depth and the level of anesthesia cannot easily control,
- Ease of occurrence of anesthetics overdose.

A. Barbiturates anesthetics

- It includes: Amobarbital, Methohexital, Thiamylal and **Thiopental**
- Can be Long acting (Phenobarbital), Short acting (Pentobarbital), and Ultra-short acting (Thiopental)

Mechanism of action

- Potentiate the effect of GABA neurotransmitter leading to depression of the brain and inhibits excessive motor discharge

Pharmacological effects

- Depresses all functions of the brain,
- Selectively depresses the motor cortex,
- Depresses the respiratory center in medulla.
- Produce good muscle relaxation.
- Decreases blood pressure and heart rate,
- Induce microsomal enzymes and increase the metabolic rate of the animal.

Clinical uses

- Sedative, Hypnotic, Anticonvulsant, Anesthetics, In case of pruritus to control itching.

B. Non-barbiturates anesthetics

Ketamine

Mechanism of action:

- Antagonizes aspartate receptors (N-methyl-D-aspartate (NMDA) receptors) leading to depression of the brain.

- It is given I.M., I.V. and S.C. It produces stage I and II anesthesia with weak muscle relaxation therefore, it is usually combined with α_2 receptor agonists like xylazine, medetomidine and detomidine.
- The analgesia is good and body reflexes like swallowing are maintained during ketamine anesthesia.

Contraindications: 1. abdominal surgery. 2. Caesarian section (CS). 3. Liver and kidney diseases. 4. Head injuries because ketamine increases the pressure of cerebrospinal fluid (CSF).

B. Local Anesthetics

Mechanism of action of local anesthetics:

- Local anesthetics prevent the local pain by blocking the sodium channel and reversibly prevent the depolarization of the nervous tissue.
- It is injected at the site of action and it is called nerve block when it is injected around or close to peripheral nerve or nerve plexuses.
- It is called epidural anesthesia when injected into the spinal cord spaces and do not interfere with the vital or local functions of the body.
- The absorption of local anesthetics can be delayed by giving epinephrine which causes vasoconstriction that prolonged the duration of anesthetic action

Types of Local Anesthetics:

E.g. Procaine, Tetracaine. Lidocaine and Etidocaine

Side effects: Hypotension, Urinary retention, Difficulties with motor muscles, Nausea and vomiting. Seizure

Clinical uses: Relief of local pain, Tail amputation in animals, Insect bites, Minor burns, Local itching

Factors Influencing Anesthetic Activity of Local Anesthetic

- Physicochemical properties (lipid solubility, protein binding, and pKa)
- Dosage of local anesthetic solutions

Common Agents Used for Local Anesthesia

Lidocaine

- The most commonly used and Very lipid soluble
- Has a more rapid onset
- Longer duration of action.
- Administration
 - ✓ Parenterally (with or without epinephrine) or
 - ✓ Topically either by itself or in combination.
- Extensively metabolized in the liver.
- Systemic toxicities (CNS changes).

2. 5. Veterinary Drugs and Chemicals for Various Diseases

2. 5. 1. Veterinary drugs for bacterial disease

- Here are some major bacterial diseases in animals and their treatments:

A. Actinobacillosis (Wooden tongue):

- This is a chronic infectious disease chiefly of cattle but occasionally of sheep.

Treatment:

- Intravenous injection of Sodium Iodide should be given and one or two injections in many cases make recovery complete.
- The effect of treatment may be enhanced by painting the lesions by Tine.
- Iodine and glycerine and by injection of Lugol's solution (10 ml to 15 ml) into the growth.

B. Actinomycosis:

Treatment:

- Penicillin or Streptomycin is quite effective.
- Sulpha drugs may also be used.
- In addition, surgical treatment should be resorted to where possible.

C. Tetanus

- A disease that can affect animals, caused by the neurotoxin produced by the bacterium *Clostridium tetani*.

Treatment

- Primarily supportive and includes thorough flushing of the wound, along with the injection of tetanus antitoxin.
- Tetanus toxoid is given for active immunization¹.
- Antibiotics and good supportive therapy can be effective.
- For farm animals like cattle, the treatment procedures include
 - ✓ Cleaning of the wound, exposure to air, flashing with Hydrogen peroxide, and applying penicillin locally and by injection.
 - ✓ Anti-tetanus serum (ATS) may also be given, which will be effective up to 8 hours after injection.
- It's important to note that all animals that have recovered from tetanus should be regularly vaccinated.
- Animals surviving tetanus do not build a good immunity and should be vaccinated with tetanus toxoid.

D. Black Quarter (Black leg)

The antibiotics that can be used to treat Black Quarter (Black leg) in cattle are:

Penicillin:

- This is the antibiotic of choice for Black Quarter.
- It should be administered in large doses, with half of the dose injected directly into the affected muscle and the other half intramuscularly.
- The treatment may be effective if given within 12 hours of the appearance of clinical symptoms.

Oxytetracycline:

- This is an alternative antibiotic that can be injected intramuscularly and into the affected muscles.

Cephalosporin:

- This is a third generation antibiotic that can also be used to treat Black Quarter.
- The dosage and frequency of these antibiotics may vary depending on the weight, age, and condition of the animal.

Anthrax:

- A highly infectious and fatal disease of cattle, caused by a relatively large spore-forming rectangular shaped bacterium called *Bacillus anthracis*.

Some of the antibiotics that can be used to treat anthrax are:

Ciprofloxacin:

- This is a broad-spectrum antibiotic that can kill anthrax bacteria and prevent them from growing. It can be taken orally or intravenously. It is usually given for 60 days to prevent anthrax after exposure.

Doxycycline:

- This is another broad-spectrum antibiotic that can be used to treat anthrax. It can also be taken orally or intravenously. It is usually given for 60 days to prevent anthrax after exposure.

Penicillin:

- This is the oldest and most widely used antibiotic for anthrax.
- It can be injected into the muscle or vein.
- It is effective against cutaneous anthrax, but may not work well against inhalation or gastrointestinal anthrax³⁴.

Oxytetracycline:

- This is an alternative antibiotic that can be used to treat anthrax.
- It can be injected into the muscle or vein. It may have fewer side effects than ciprofloxacin or doxycycline.

Cephalosporin:

- This is a newer generation of antibiotic that can also be used to treat anthrax.
- It can be injected into the vein. It may be more effective against inhalation anthrax than penicillin.

Bovine tuberculosis (bTB)

- A chronic bacterial disease of animals caused by members of the Mycobacterium tuberculosis complex, primarily by M. bovis.
- It can affect the lungs, lymph nodes, and other parts of the body in cattle and other animals, as well as humans.

The treatment of bTB depends on the species, the stage of the disease, and the availability of antibiotics. Some of the antibiotics that can be used to treat bTB are:

Penicillin:

This is the oldest and most widely used antibiotic for bTB.

- It can be injected into the muscle or vein.
- It is effective against cutaneous bTB, but may not work well against inhalation or gastrointestinal bTB.

Ciprofloxacin:

- This is a broad-spectrum antibiotic that can kill bTB bacteria and prevent them from growing.
- It can be taken orally or intravenously. It is usually given for 60 days to prevent bTB after exposure.

Doxycycline:

- This is another broad-spectrum antibiotic that can be used to treat bTB.
- It can also be taken orally or intravenously. It is usually given for 60 days to prevent bTB after exposure.

Oxytetracycline:

- This is an alternative antibiotic that can be used to treat bTB.
- It can be injected into the muscle or vein. It may have fewer side effects than ciprofloxacin or doxycycline.

Cephalosporin:

- This is a newer generation of antibiotic that can also be used to treat bTB.
- It can be injected into the vein. It may be more effective against inhalation bTB than penicillin.

Note:

- Prevention of bTB in animals and humans can be achieved by vaccination, avoiding contact with infected animals or their products, and pasteurizing milk and dairy products.
- Antimicrobial treatment of infected animals is rarely attempted because of the doses and duration of treatment that would be required, high cost of medications, and interference with the primary goal of eliminating the disease, and potential risk of developing resistance.
- The treatment consists of taking antibiotics for a period of 6 to 12 months. However, there's currently no vaccine or effective treatment.

Brucellosis

- A bacterial infection that spreads from infected animals or unpasteurized animal products to humans¹. The symptoms may include fever, joint pain, and fatigue¹.
- It is caused by different species of the genus *Brucella*, which can be transmitted through contact with infected animals or their products, such as unpasteurized milk or cheese.
- The treatment of brucellosis in animals depends on the type and severity of the infection, as well as the species, condition, and medical history of the animal.
- The main treatment is antibiotics, which can kill the bacteria and prevent complications.
- However, the treatment may take several weeks or months, and the infection may relapse or become chronic in some cases.

Some of the antibiotics that can be used to treat brucellosis in animals are:

Doxycycline:

This is a broad-spectrum antibiotic that can be taken orally or intravenously.

It is usually given in combination with another antibiotic, such as rifampin, streptomycin, or gentamicin, for at least 6 to 8 weeks.

Rifampin:

- This is another broad-spectrum antibiotic that can be taken orally or intravenously.
- It is usually given in combination with doxycycline or streptomycin for at least 6 to 8 weeks.

Streptomycin:

- This is an aminoglycoside antibiotic that can be injected into the muscle or vein.
- It is usually given in combination with doxycycline or rifampin for at least 6 to 8 weeks.

Tetracycline:

- This is an alternative antibiotic that can be taken orally or intravenously.
- It is similar to doxycycline, but it may have more side effects and interactions with other drugs.

Ciprofloxacin:

- This is a fluoroquinolone antibiotic that can be taken orally or intravenously.
- It can be used as a prophylaxis after exposure to brucellosis or as a treatment for mild cases.

Penicillin:

- This is the oldest and most widely used antibiotic for brucellosis.
- It can be injected into the muscle or vein. It is effective against cutaneous brucellosis, but it may not work well against inhalation or gastrointestinal brucellosis.

Note:

- The dosage and frequency of these antibiotics may vary depending on the weight, age, and condition of the animal. Therefore, it is advisable to consult a veterinarian before starting any treatment.
- Prevention of brucellosis in animals can be achieved by vaccination, avoiding contact with infected animals or their products, pasteurizing milk and dairy products, and wearing protective gloves and clothing when handling animals or their tissues.

Botulism:

- A serious and potentially fatal illness caused by a toxin produced by the bacterium *Clostridium botulinum*.
- The treatment of botulism depends on the type and severity of the infection, as well as the patient's condition and medical history.
- The main treatment is antitoxin, which is a drug that neutralizes the toxin and prevents it from causing more damage.
- Antitoxin does not heal the damage that has already been done, but it can reduce the risk of complications and death.
- Antitoxin should be given as soon as possible after diagnosis, preferably within 24 hours.
- Antibiotics are not usually used to treat botulism, except for wound botulism.
- This is because antibiotics can speed up the release of toxins from the bacteria, making the condition worse.
- Antibiotics may also have negative effects on the patient's health, such as causing diarrhea, reducing immunity, or selecting for resistant bacteria.
- Therefore, antibiotics should be used with caution and under medical supervision.

Some of the antibiotics that may be used to treat wound botulism are:

Penicillin:

- This is the oldest and most widely used antibiotic for wound botulism.

- It can be injected into the muscle or vein. It is effective against cutaneous botulism, but it may not work well against inhalation or gastrointestinal botulism.

Ciprofloxacin:

- This is a broad-spectrum antibiotic that can kill botulism bacteria and prevent them from growing. It can be taken orally or intravenously.
- It is usually given for 60 days to prevent botulism after exposure.

Doxycycline:

- This is another broad-spectrum antibiotic that can be used to treat wound botulism.
- It can also be taken orally or intravenously. It is usually given for 60 days to prevent botulism after exposure.

Tetracycline:

- This is an alternative antibiotic that can be used to treat wound botulism.
- It can be taken orally or intravenously.
- It is similar to doxycycline, but it may have more side effects and interactions with other drugs.

Cephalosporin:

- This is a newer generation of antibiotic that can also be used to treat wound botulism.
- It can be injected into the vein.
- It may be more effective against inhalation botulism than penicillin³.
- The dosage and frequency of these antibiotics may vary depending on the weight, age, and condition of the patient. Therefore, it is advisable to consult a doctor before starting any treatment.
- Supportive care, such as breathing assistance, nutrition, hydration, pain relief, and rehabilitation therapy.

Contagious bovine pleuro pneumonia (CBPP)

- A serious respiratory disease of cattle caused by a bacterium called *Mycoplasma mycoides* subsp. *mycoides* (MmmSC).
- The treatment of CBPP is challenging and often ineffective, as the bacterium is resistant to many antibiotics and can persist in the lungs of recovered animals.

Some of the possible treatments for CBPP are:

- Antibiotics: Some antibiotics, such as tetracyclines, fluoroquinolones, and macrolides, may have some activity against, but they are not very effective in curing the disease or preventing its spread.
- Antibiotics may be given orally or intravenously, depending on the availability and the condition of the animal. However, antibiotics may also have negative effects on the animal's health, such as causing diarrhea, reducing immunity, or selecting for resistant bacteria.

Table 2.1 Summary of bactericidal Drug with their use in animals

Bactericidal Drug	Use in Animals
Cephalosporins	Effective against many types of bacteria and commonly used in the treatment of respiratory, skin, and urinary tract infections in animals.
Aminoglycosides	Used to treat severe bacterial infections in animals, particularly Gram-negative infections such as pneumonia, septicemia, and urinary tract infections.
Fluoroquinolones	Effective against a broad spectrum of bacteria and used to treat respiratory, urinary tract, gastrointestinal, and skin infections in animals.
Vancomycin	Primarily used to treat severe Gram-positive bacterial infections in animals, such as methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) infections.
Metronidazole	Effective against anaerobic bacteria and protozoal infections in animals, commonly used in the treatment of gastrointestinal and periodontal infections.
Rifampin	Used in combination therapy to treat Mycobacterial infections, including tuberculosis and Johne's disease in animals.
Penicillin	Used to treat a wide range of bacterial infections in various animal species, including respiratory infections, skin infections, and urinary tract infections.

Table 2.2: Summary of bacteriostatic drug with their use in animals

Bacteriostatic Drug	Use in Animals
Tetracycline's	Widely used in animals to treat respiratory, urinary tract, gastrointestinal, and skin infections caused by susceptible bacteria.
Macrolides	Used to treat respiratory, skin, and gastrointestinal infections in animals. Examples include erythromycin, azithromycin, and tylosin.

Sulfonamides	Used to treat a variety of bacterial infections in animals, including respiratory, urinary tract, and gastrointestinal infections.
Trimethoprim	Often combined with sulfonamides to create potent combinations for the treatment of urinary tract and respiratory infections in animals.
Chloramphenicol	Used as a bacteriostatic drug in animals to treat a range of bacterial infections, including respiratory and gastrointestinal infections.
Lincosamides	Used primarily in veterinary medicine to treat skin and soft tissue infections in animals caused by susceptible bacteria.
Florfenicol	Effective against respiratory and gastrointestinal infections in animals, particularly in cases where other antibiotics may be ineffective.

2. 5. 2. Veterinary drugs for fungal diseases

Table 2.3: Summary of antifungal drug with their use in animals

Antifungal Drug	Use in Animals
Azoles (e.g., Fluconazole)	Used to treat systemic fungal infections in animals, including infections caused by Candida and Aspergillus species.
Polyenes (e.g., Amphotericin B)	Effective against severe systemic fungal infections in animals, including those caused by Cryptococcus and Histoplasma.
Griseofulvin	Used primarily to treat dermatophyte infections (ringworm) in animals.
Allylamines (e.g., Terbinafine)	Effective against dermatophyte infections in animals, including ringworm.
Nystatin	Used to treat fungal infections of the gastrointestinal tract in animals, such as candidiasis.
Itraconazole	Effective against systemic and localized fungal infections in animals, including Aspergillus and dermatophyte infections.
Ketoconazole	Used to treat systemic and superficial fungal infections in animals.

2. 6. Veterinary Drugs for Parasitic Diseases

Table 2.4: Summary of anthelmintic drug with their use in animals

Anthelmintic Drug	Use in Animals
Benzimidazoles	Used to treat various internal parasites in animals, including roundworms, hookworms, whipworms, and some tapeworms.
Macrocyclic Lactones	Effective against a broad range of internal parasites in animals, including roundworms, hookworms, heartworms, and certain external parasites.
Imidazothiazoles	Used to control and treat gastrointestinal nematodes, lungworms, and some external parasites in animals.
Tetrahydropyrimidines	Primarily used to treat gastrointestinal nematodes, such as roundworms and hookworms, in animals.
Salicylanilides	Used to treat or prevent cestode (tapeworm) infections in animals.
Avermectins	Broad-spectrum anthelmintics used to treat gastrointestinal and lungworm infections, as well as external parasites like mites and lice in animals.
Piperazine	Effective against roundworm infections in animals.
Isoquinoline Derivatives	Used to treat cestode (tapeworm) infections in animals.
Triclabendazole	Effective against liver fluke infections in animals, particularly <i>Fasciola hepatica</i> and <i>Fasciola gigantica</i> .
Pyrantel	Used to treat gastrointestinal nematode infections in animals, including roundworms and hookworms.
Levamisole	Effective against various nematodes in animals, including gastrointestinal worms and lungworms.
Praziquantel	Widely used to treat various trematode infections in animals, including liver flukes, blood flukes, and intestinal flukes.
Fenbendazole	Used to treat certain cestode infections in animals, including <i>Taenia</i> species and <i>Echinococcus multilocularis</i> .
Niclosamide	Primarily used to treat cestode infections in animals, including <i>Taenia</i> species and <i>Echinococcus multilocularis</i> .

Protozoa diseases

Coccidiosis

- a parasitic disease caused by the *Coccidia* protozoa. The disease can affect various animals, including dogs, cats, and cattle.

Here are some treatments for coccidiosis:

- Sulfadimethoxine

Bovine babesiosis

- A tick-borne disease of cattle caused by protozoan parasites of the genus *Babesia*.

The recommended treatments for bovine babesiosis are:

- Imidocarb Dipropionate:
 - ✓ This is administered via intramuscular or subcutaneous injection, at a dosage of 1-3 mg/kg.
 - ✓ It is effective for dogs, horses, and cattle.
- Diminazene Aceturate (Berenil):
 - ✓ This is injected intramuscularly at a dosage of 3-5 mg/kg.

Bovine trypanosomiasis,

- A disease caused by protozoa of the genus *Trypanosoma*, affects all domesticated animals.

The disease is currently controlled using a combination of methods:

- Trypanotolerant Host: Using animals that are naturally resistant to the disease.
- Vector Control: Managing the tsetse fly population, this is responsible for transmitting the disease.
- Trypanocidal Drugs: Several drugs can be used for treatment.

2. 6. Environmental health policies and legislations to wastes disposal

- Environmental health policies and legislations regarding the disposal of drug and chemical wastes vary by country and jurisdiction.
- These policies aim to ensure the safe and proper handling, storage, transportation, and disposal of hazardous waste to protect human health and the environment.

Some general concepts and practices that are commonly followed in many countries:

- ✓ **Hazardous Waste Classification:** Environmental health regulations typically classify certain drugs and chemicals as hazardous waste due to their potential risks to human health and the environment. These wastes may include expired or unused pharmaceuticals, chemical reagents, contaminated containers, and other related materials.

- ✓ **Waste Segregation and Labeling:** Proper segregation and labeling of drug and chemical waste are essential for safe handling and disposal. Waste should be segregated based on its properties and compatibility to prevent cross-contamination. Clear labeling ensures that waste handlers are aware of the potential hazards and appropriate precautions to take.
- ✓ **Storage and Containment:** Hazardous drug and chemical waste should be stored in appropriate containers that are designed to prevent leaks, spills, or releases. Storage areas should be secure, well-ventilated, and equipped with measures to contain potential spills or leaks.
- ✓ **Transportation:** Transportation of hazardous waste should comply with local regulations and guidelines. It often involves using licensed waste management companies or specialized transport services that are trained in handling and transporting hazardous materials safely.
- ✓ **Treatment and Disposal:** Hazardous drug and chemical waste may require specific treatment processes before disposal. Treatment methods can include incineration, chemical neutralization, or physical processes to render the waste less hazardous. Final disposal should be in compliance with applicable regulations, which may involve landfilling in designated facilities or other approved methods.
- ✓ **Reporting and Record-Keeping:** Regulatory requirements often include reporting and record-keeping obligations for entities generating, handling, or disposing of drug and chemical waste. This helps ensure transparency, accountability, and regulatory compliance.
- ✓ **Monitoring and Enforcement:** Environmental health agencies may conduct inspections, audits, and monitoring activities to ensure compliance with waste disposal regulations. Non-compliance can result in penalties, fines, or legal actions.

Self-check 2	Written test
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Name.....ID..... Date.....

Directions: Answer all the questions listed below.

Part I: Write “True” if the statement is correct or “False” if the statement is incorrect (10 Points)

1. Quinidine Sulphate is primarily used to treat or prevent irregular heartbeats such as atrial fibrillation.
2. Procainamide Hydrochloride is mainly used for infiltration anesthesia, peripheral nerve block, and spinal block.
3. Verapamil is used to prevent chest pain caused by angina and treat irregular heart rhythms.
4. Amiodarone is used to treat and prevent certain types of serious, life-threatening ventricular arrhythmias.
5. Sotalol is used to treat atrial fibrillation and other conditions that cause an irregular heartbeat.
6. Propranolol is primarily used to treat high blood pressure, angina, and arrhythmias.
7. Captopril is an angiotensin-converting enzyme (ACE) inhibitor used to relax blood vessels and lower blood pressure.
8. Ramipril is a calcium channel blocker used to treat high blood pressure.
9. Isosorbide Dinitrate is used to increase blood flow to the heart and treat angina pectoris.
10. Digoxin is an antianginal agent used to treat chest pain caused by reduced blood flow to the heart.

Part II: Match column “A” with appropriate answer from column “B”

Part III. Choose the best answer for the following question (7 Points)

1. Which of the following drugs is commonly used as an appetite stimulant for cachexic and anorexic animals, as well as RF dogs?

A. Metoclopramide

B. Diazepam

C. Apomorphine

D. Xylazine

2. Which of the following drugs is an emetic used primarily for induction of vomiting in dogs and is considered the emetic of choice for dogs?
 - A. Apomorphine
 - B. Xylazine
 - C. Hydrogen peroxide
 - D. Metoclopramide
3. Which drug is commonly used orally to induce vomiting in dogs, cats, pigs, and ferrets?
 - A. Apomorphine
 - B. Xylazine
 - C. Hydrogen peroxide
 - D. Metoclopramide
4. Which of the following is not a contraindication for using emetic drugs?
 - A. Acid or alkali poisoning
 - B. CNS stimulant poisoning
 - C. Kerosene (Petroleum) poisoning
 - D. Unconscious patients
5. Which class of drugs acts centrally by blocking dopamine receptors in the CRTZ and is commonly used as antiemetics?
 - A. Phenothiazine derivatives
 - B. Procainamide derivatives
 - C. Antihistamines
 - D. Serotonin receptor antagonists
6. Which antiemetic drug is more effective in dogs than in cats and is commonly used in veterinary medicine?
 - A. Chlorpromazine
 - B. Metoclopramide
 - C. Trimethobenzamide HCl
 - D. Ondansetron
7. Which class of drugs block serotonin receptors (5-HT₃) and are known for their antiemetic activity?
 - A. Phenothiazine derivatives
 - B. Procainamide derivatives
 - C. Antihistamines
 - D. Serotonin receptor antagonists

Part IV: Short Answer Questions (7 points)

1. What is the purpose of veterinary drugs?
2. How are acaricides used in animals?
3. What is the role of anthelmintic drugs in treating animals?
4. Give an example of an antiprotozoal drug used in veterinary medicine.
5. How are antifungal drugs used to treat animals?
6. Name some food additives used in the preservation and processing of animal food.
7. How do antimicrobial agents contribute to growth promotion in food animals?

Notice: Scoring rates of above **12** is satisfactory and less than that is not satisfactory

LG #3	LO #3-Identify and administer biological fluids
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Instruction sheet 3

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- Fluids therapy for injured/ diseased animals
- Dose and frequency of biological fluid

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Identify and prepare appropriate fluids for therapy of injured/ diseased animals.
- Calculated dose and frequency of biological fluid.
- Administer fluid therapy for controlling metabolic diseases and prescribe as supplement for needy animals according to organizational treatment protocol.
- Kept record in and reported to the supervisor.

Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

Information Sheet 3

3. 1. Fluids Therapy for Injured/ Diseased Animals

Fluid therapy in clinical medicine is used to fulfill the following objectives:

- To replace dehydration deficits,
- To maintain normal hydration,
- To replace essential electrolytes and nutrients, and
- To serve as a vehicle for the infusions of certain intravenous medications.

A. Dextrose

- It is used in veterinary medicine for various purposes:
- Used in fluid therapy for critically ill dogs and cats.
- It helps to replace dehydration deficits, maintain normal hydration, replace essential electrolytes and nutrients, and serve as a vehicle for the infusions of certain intravenous medications.
- Hypertonic solutions like dextrose 5% in 0.9% saline can be used as a partial maintenance solution once the patient is completely rehydrated.

Dextrose used for the following condition:

I. Hypoglycemia Treatment:

- May be given intravenously (by IV) to help raise blood sugar levels in animals with diabetes mellitus who are having a hypoglycemic reaction.
- It is also used as an emergency treatment for low blood sugar levels in animals that cannot eat or drink.

II. Energy Source during Surgery:

- May be given intravenously (by IV) to help provide energy for animals during surgery.

III. Ringer lactate

- Also known as Lactated Ringer's Injection, is used in veterinary medicine for various purposes:

IV. Fluid Therapy:

- Ringer's lactate is used in fluid therapy for critically ill animals.
- It helps to replace dehydration deficits, maintain normal hydration, replace essential electrolytes and nutrients, and serve as a vehicle for the infusions of certain intravenous medications.

V. Correction of Electrolyte Depletion, Metabolic Acidosis, and Dehydration:

- Ringer's lactate is indicated for the correction of electrolyte depletion, metabolic acidosis and dehydration of cattle, calves, horses, sheep and swine.

B. Ringer's lactate:

Alkalinizing Agent: Ringer's lactate is also used as an alkalinizing agent.

- Ringer's lactate may be injected intravenously, subcutaneously or intraperitoneally (except in horses) using strict aseptic technique.
- However, it should not be administered to animals with inadequate renal function¹.
- It's also important to note that the solution should be warmed to body temperature prior to administration and administered at a slow rate.

I. Saline solution

- Saline solution, also known as Sterile Saline Solution, is used in veterinary medicine for various purposes:

II. Fluid Therapy:

- Saline solution is used in fluid therapy for critically ill animals.
- It helps to replace dehydration deficits, maintain normal hydration, replace essential electrolytes and nutrients, and serve as a vehicle for the infusions of certain intravenous medications.

III. Correction of Electrolyte Depletion and Dehydration:

- Saline solution is indicated for the correction of electrolyte depletion and dehydration which may become depleted in many diseases¹.
- Because this solution is isotonic with body fluids, it may also be used as a solvent or diluent, for antibiotics and other pharmaceuticals and biologicals where compatible.

IV. Washing Mucous Membranes and Other Tissue Surfaces:

- Saline solution can be used for washing mucous membranes and other tissue surfaces¹.

Note: Saline solution may be injected intravenously or subcutaneously using strict aseptic technique.

C. Calcium borogluconate

- It is used in veterinary medicine for various purposes:

I. Treatment of Hypocalcaemia:

- Calcium Borogluconate is used as a source of calcium for cattle, sheep, and swine, to aid in the treatment of hypocalcaemia, e.g., milk fever in cattle.

II: Increase Plasma Concentrations of Calcium and Magnesium:

- On parenteral administration, Calcium Borogluconate rapidly increases plasma concentrations of calcium and magnesium.
- This is effective in the treatment of hypocalcemia with associated hypomagnesaemia.

Dosage and Administration:

- It should be administered slowly by intravenous or subcutaneous injection.
- Intravenous administration is recommended for the treatment of acute conditions.
- When administered subcutaneously, the dose should be divided over several sites.
- The solution should be warmed to body temperature prior to use.
- Dosage may be repeated after several hours if required or as recommended by a veterinarian.
- The dosage for adult cattle is 250 to 500 mL and for adult sheep and swine is 50 to 125 mL or as recommended by a veterinarian.

3. 2. Dose and Frequency of Biological Fluid

A: Calcium Borogluconate:

- Used for the treatment of hypocalcemia or milk fever, in cattle.
- The recommended dosage for adult cattle is 250 to 500 mL.
- It should be administered slowly by intravenous or subcutaneous injection.
- If administered subcutaneously, the dose should be divided over several sites

- The solution should be warmed to body temperature prior to use.
- The dosage may be repeated after several hours if required or as recommended by a veterinarian.

B: Ringer's lactate (Lactated Ringer's Injection)

- Used for the correction of electrolyte depletion, metabolic acidosis, and dehydration in animals.

The recommended dosage and frequency for different animals are as follows:

- **Cattle and Horses:** 2 to 5 mL per pound of body weight depending on size and condition of the animal, repeated 1 to 3 times daily or as needed.
- Swine and Sheep: 2 to 5 mL per pound of body weight depending on size and condition of the animal, repeated 1 to 3 times daily or as needed.
- The solution may be injected intravenously, subcutaneously, or intraperitoneally (except in horses) using strict aseptic technique.
- If administered subcutaneously, the dosage should be divided into several sites of injection and massage points of injection to aid in absorption and help prevent inflammation and/or sloughing.
- The solution should be warmed to body temperature prior to administration and administered at a slow rate.

C. Dextrose (glucose)

- A simple sugar that can be used in veterinary medicine as a source of carbohydrates for animals.
- It is often used in cases of hypoglycemia or low blood sugar.

Some general guidelines for dextrose administration in animals:

- For life-saving bolus therapy in a hypoglycemic crisis, supplementation should be implemented quickly! Use this shortcut for those crisis situations: 0.5–1.5 mL/kg of 50% dextrose diluted 1:42. This can be followed by dextrose in solution as noted below.
- To make a 2.5% solution, add 50mL of 50% dextrose (or 25g dextrose) to a 1L bag of fluids.
- To make a 5.0% solution, add 100mL of 50% dextrose (or 50g dextrose) to a 1L bag of fluids.

- For oral administration in adult cattle, the usual recommended dose is 16.9 fl. oz. (500 mL) to 33.8 fl. oz. (1000 mL) depending on size and condition.

Self-check 3	Written test
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Name..... ID..... Date.....

Directions: Answer all the questions listed below.

Part I: Write “True” if the statement is correct or “False” if the statement is incorrect (7 Points)

1. Dextrose administered slowly by intravenous or subcutaneous injection.
2. Lactated Ringer’s Injection used for the correction of electrolyte depletion, and dehydration in animals.
3. Effective in the treatment of hypocalcaemia with associated hypomagnesaemia
4. Dextrose can be used in fluid therapy for critically ill dogs and cats.
5. Saline solution can be used for washing mucous membranes and other tissue surfaces.
6. Calcium borogluconate is used as a source of calcium for the treatment of hypocalcaemia, such as milk fever in cattle.
7. Intravenous administration is recommended for the treatment of acute conditions with calcium borogluconate.

Part II. Choose the best answer for the following question (6 Points)

1. Which of the following objectives does fluid therapy in clinical medicine aim to fulfil?
 - a) Replacement of essential electrolytes and nutrients
 - b) Maintenance of normal hydration
 - c) Reversal of hypoglycemia
 - d) All of the above
2. Which intravenous medication can be used as a vehicle in fluid therapy?
 - a) Dextrose
 - b) Ringer lactate

- c) Saline solution
- d) Calcium borogluconate
3. In fluid therapy for critically ill dogs and cats, which solution can be used to replace dehydration deficits and maintain normal hydration?
 - a) Dextrose
 - b) Ringer lactate
 - c) Saline solution
 - d) Calcium borogluconate
4. Which solution is indicated for the correction of electrolyte depletion, metabolic acidosis, and dehydration in cattle, horses, and sheep?
 - a) Dextrose
 - b) Ringer lactate
 - c) Saline solution
 - d) Calcium borogluconate
5. What is the purpose of using saline solution in fluid therapy for critically ill animals?
 - a) Replacing dehydration deficits
 - b) Maintaining normal hydration
 - c) Replacing essential electrolytes and nutrients
 - d) All of the above
6. Which solution is used as a source of calcium for the treatment of hypocalcemia, such as milk fever in cattle?
 - a) Dextrose
 - b) Ringer lactate
 - c) Saline solution
 - d) Calcium borogluconate

LG #4	LO #4-Store and dispense veterinary drugs and chemicals
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Instruction sheet 4

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- PPE and equipment for storing and dispensing drugs
- Handling and dispensing of Pharmaceutical Products

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Use PPE clothing and equipment relevant in storing and dispensing drugs/chemical in accordance with organizational policies and procedures.
- Store medicines and chemicals in accordance with the manufacturer's specification or datasheet.
- Protected medicines from environmental conditions that may damage or degrade them such as light, temperature and humidity according to the organizational guideline.
- Use organizational operating guidelines in the handling and dispensing of drugs.

Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

Information Sheet 4

4. 1. PPE and Equipment for Storing and Dispensing Drugs

- When storing and dispensing drugs, it's important to have the appropriate personal protective equipment (PPE) and other necessary equipment.

Some recommendations:

A. Personal Protective Equipment (PPE):

- PPE should be provided and used wherever risks cannot be adequately controlled in other ways.
- PPE for handling hazardous drugs in healthcare settings includes chemotherapy gloves and gowns. <https://www.hse.gov.uk/healthservices/safe-use-cytotoxic-drugs.htm>

B. Equipment:

- For pharmacies, you would need an autoclave, mortar and pestle, a compounding hood, spatulas, vials, scales, flasks, needles and syringes, and IV materials.
- Use of engineering controls, such as ventilated hoods and enclosures.

C. Administrative Controls:

- Establish safe handling policies, training, and routine training reviews for potentially exposed individuals.

Note: Remember to always follow the guidelines provided by health and safety authorities and consult with a professional for specific requirements in your context.

4. 2. Handling and dispensing of Pharmaceutical Products

- Handling and dispensing of pharmaceutical products involve several important steps to ensure the safety, efficacy, and quality of the medicines.

Some key points:

I. Obtaining Medicines: The process starts with obtaining the medicines from a reliable source, ensuring their quality and authenticity.

II. Transport: Medicines should be transported under appropriate conditions to maintain their quality.

III. Receipt and Storage: Upon receipt, the medicines should be checked for any damage or discrepancies. They should be stored under conditions specified by the manufacturer to maintain their quality.

IV. Dispensing: Dispensing includes checking the validity of the prescription, the appropriateness of the medicine for an individual patient, assembly of the product, labelling in accordance with legal requirements, and providing advice to the patient on safe and effective use of these products.

V. Removal or Disposal: Expired or damaged medicines should be disposed of safely to prevent any harm. It's important to have appropriate personal protective equipment (PPE) for handling medicines.

Self-check 4	Written test
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Name..... ID..... Date.....

Directions: Answer all the questions listed below.

Part I: Write “True” if the statement is correct or “False” if the statement is incorrect (5 Points)

1. Personal Protective Equipment (PPE) should be used when handling hazardous drugs in healthcare settings.
2. PPE for handling hazardous drugs includes chemotherapy gloves and gowns.
3. Autoclave, mortar and pestle, and compounding hood are examples of equipment needed in a pharmacy for storing and dispensing drugs.
4. Engineering controls, such as ventilated hoods and enclosures, should be used to control the handling of hazardous drugs.
5. Safe handling policies, training, and routine training reviews should be established for individuals potentially exposed to hazardous drugs.

Part II. Choose the best answer for the following question (5 Points)

1. What is the purpose of personal protective equipment (PPE) when handling hazardous drugs in healthcare settings?
 - A. To enhance the taste and smell of the drugs
 - B. To ensure patient comfort during drug administration
 - C. To protect healthcare workers from exposure to harmful substances
 - D. To improve the therapeutic effect of the drugs
2. What are some examples of equipment needed in a pharmacy for storing and dispensing drugs?
 - A. Autoclave, spatulas, and vials
 - B. Microscope, stethoscope, and blood pressure cuff
 - C. Computer, printer, and barcode scanner
 - D. Refrigerator, microwave, and coffee machine
3. What are engineering controls in the context of handling hazardous drugs?
 - A. Tools and equipment used for drug compounding

- B. Safety measures implemented in the pharmacy workflow
 - C. Techniques for conducting quality control tests on drugs
 - D. Ventilated hoods and enclosures to minimize exposure to hazardous substances
4. What are administrative controls in the context of safe handling of hazardous drugs?
- A. Policies, training, and routine training reviews for potentially exposed individuals
 - B. Physical barriers and protective barriers in the pharmacy
 - C. Guidelines for proper disposal of expired drugs
 - D. Techniques for labeling and packaging pharmaceutical products
5. What are the key steps involved in the handling and dispensing of pharmaceutical products?
- A. Manufacturing, marketing, and distribution
 - B. Prescription verification, product assembly, and patient counseling
 - C. Quality control, packaging, and labeling
 - D. Research and development, clinical trials, and regulatory approval

Part III: Short Answer Questions (6 points)

1. List the recommendation when storing and dispensing drugs (3 points)?
2. List steps to Handling and dispensing of pharmaceutical products (3 points)?

Notice: Scoring rates of above **8** is satisfactory and less than that is not satisfactory

LG #5

LO #5-Apply stock control and clinic security

Instruction Sheet 5

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- Principles of veterinary medical ethics
- Stock control and inventory of veterinary drugs
- Veterinary clinic security measures

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Take regular inventory of veterinary supplies and medicines according to clinic policies.
- Order medicines and supplies under direction of the duty veterinarian and qualified veterinary nurse.
- Check clinic stocks as they received against quantities ordered and priced.
- Handle and store drugs securely in accordance with clinic and legislative requirements; legislative requirements relating to the handling, storage and security of drugs, including dangerous drugs should be considered
- Use clinic security measures in accordance with clinic procedures.

Learning Instructions:

10. Read the specific objectives of this Learning Guide.
11. Follow the instructions described below.
12. Read the information written in the information Sheets
13. Accomplish the Self-checks

Information Sheet 5

5. 1. Principles of Veterinary Medical Ethics

- The Principles of Veterinary Medical Ethics (PVME) are a set of guidelines established by the American Veterinary Medical Association (AVMA) to uphold the dignity and professionalism of the veterinary profession.

Key principles:

1. A veterinarian shall be influenced only by the welfare of the patient, the needs of the client, the safety of the public, and the need to uphold the public trust vested in the veterinary profession, and shall avoid conflicts of interest or the appearance thereof.
2. A veterinarian shall provide competent veterinary medical clinical care under the terms of a veterinarian-client-patient relationship (VCPR), with compassion and respect for animal welfare and human health.
3. A veterinarian shall uphold the standards of professionalism, be honest in all professional interactions, and report veterinarians who are deficient in character or competence to the appropriate entities.
4. A veterinarian shall respect the law and also recognize a responsibility to seek changes to laws and regulations which are contrary to the best interests of the patient and public health.
5. A veterinarian shall respect the rights of clients, colleagues, and other health professionals, and shall safeguard medical information within the confines of the law.
6. A veterinarian shall continue to study, apply, and advance scientific knowledge, maintain a commitment to veterinary medical education, make relevant information available to clients, colleagues, the public, and obtain consultation or referral when indicated.
7. A veterinarian shall, in the provision of appropriate patient care, except in emergencies, be free to choose whom to serve, with whom to associate, and the environment in which to provide veterinary medical care.
8. A veterinarian shall recognize a responsibility to participate in activities contributing to the improvement of the community and the betterment of public health.

9. A veterinarian should view, evaluate, and treat all persons in any professional activity or circumstance in which they may be involved, solely as individuals on the basis of their own personal abilities, qualifications, and other relevant characteristics.

5. 2. Stock Control and Inventory of Veterinary Drugs

- Stock control and inventory of veterinary drugs is an important aspect of managing a veterinary practice.

It involves ordering, storing, dispensing, and disposing of drugs in a safe and efficient manner.

Stock control and inventory of veterinary drugs can help to:

- Ensure the availability and quality of drugs for animal health care
- Reduce the risk of drug shortages, spoilage, theft, or misuse
- Comply with legal and ethical requirements for drug handling and record keeping
- Optimize the cost and profitability of drug purchases and sales

Some of the best practices for stock control and inventory of veterinary drugs are:

- Take regular inventory of controlled drugs. Assign drug ordering and inventory tasks to different employees. Maintain accurate dispensing records. Follow DEA rules for disposal of expired controlled drugs: render them “non-retrievable” in the presence of two witnesses and document on DEA Form.
- Use an inventory management system, such as a spreadsheet or a software application, to track the quantity, location, expiration date, and cost of each drug. Update the system regularly and reconcile it with physical counts.
- Establish a minimum and maximum stock level for each drug based on the demand, turnover rate, lead time, and shelf life. Order new drugs when the stock level reaches the minimum or reorder point. Avoid overstocking or understocking drugs.
- Store drugs in a secure, clean, dry, and temperature-controlled area. Follow the manufacturer’s instructions for proper storage conditions. Separate expired, damaged, or recalled drugs from usable ones. Rotate the stock using the first-in first-out (FIFO) method.
- Price drugs according to the market value, cost of goods sold (COGS), overhead expenses, and desired profit margin. Use a markup or margin calculator to determine the optimal selling price. Adjust prices periodically based on supply and demand.

- Educate staff on the importance and procedures of stock control and inventory of veterinary drugs. Provide training on how to use the inventory management system, how to order, store, dispense, and dispose of drugs, and how to prevent and report errors or discrepancies.

5. 3. Veterinary Clinic Security Measures

- Security measures are important for any veterinary clinic, as they can protect the staff, the clients, the animals, and the property from potential threats.

Some of the common security measures that can be incorporated in a veterinary clinic are:

Lighting:

- Lighting up the exterior and interior of the clinic can deter intruders and vandals, as well as provide visibility and safety for the staff and clients.
- Lighting can also help to monitor the activity around the clinic using security cameras or alarms.

Security cameras:

- Security cameras can record and store video footage of the clinic premises, which can be useful for identifying and prosecuting any perpetrators of crime or damage.
- Security cameras can also act as a deterrent, as they signal that the clinic is monitored and protected.

Physical barriers:

- Physical barriers such as fences, gates, locks, windows, and doors can prevent unauthorized access to the clinic or certain areas within the clinic.
- Physical barriers can also protect the animals from escaping or being harmed by intruders.

Alarm systems:

- Alarm systems can alert the staff, the clients, or the authorities of any suspicious or dangerous activity in or around the clinic.
- Alarm systems can be triggered by motion sensors, door or window contacts, glass break detectors, panic buttons, or smoke detectors.

Safe storage:

- Safe storage of drugs, equipment, records, and money can prevent theft, misuse, or loss of valuable or sensitive items.

- Safe storage can include using locked cabinets, safes, drawers, or boxes, as well as keeping an inventory of the items and limiting access to authorized personnel only.

Self-check 5	Written test
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Name..... ID..... Date.....

Directions: Answer all the questions listed below.

Part I: Write “True” if the statement is correct or “False” if the statement is incorrect (5 Points)

1. The Principles of Veterinary Medical Ethics are established by the American Veterinary Medical Association (AVMA) to uphold the dignity and professionalism of the veterinary profession.
2. Stock control and inventory of veterinary drugs help ensure the availability and quality of drugs for animal health care.
3. Using an inventory management system is not necessary for tracking the quantity, location, expiration date, and cost of veterinary drugs.
4. Security cameras in a veterinary clinic can serve as a deterrent and help identify and prosecute any perpetrators of crime or damage.
5. Safe storage of drugs, equipment, records, and money in a veterinary clinic is not necessary to prevent theft, misuse, or loss of valuable items.

Part II. Choose the best answer for the following question (5 Points)

1. Which of the following is NOT a key principle of the Principles of Veterinary Medical Ethics (PVME)?
 - a) A veterinarian shall be influenced only by the welfare of the patient
 - b) A veterinarian shall respect the rights of clients, colleagues, and other health professionals
 - c) A veterinarian shall prioritize personal financial gain over patient care
 - d) A veterinarian shall continue to study and advance scientific knowledge
2. What is one of the benefits of implementing stock control and inventory of veterinary drugs?
 - a) Increasing the risk of drug shortages and spoilage
 - b) Reducing compliance with legal and ethical requirements
 - c) Optimizing the cost and profitability of drug purchases and sales
 - d) Decreasing the availability and quality of drugs for animal health care
3. Which method should be used to store drugs in a veterinary clinic?

- a) Exposing drugs to direct sunlight
 - b) Storing drugs in a temperature-controlled area
 - c) Mixing expired, damaged, and usable drugs together
 - d) Using the last-in first-out (LIFO) method for stock rotation
4. What is the purpose of using security cameras in a veterinary clinic?
- a) To provide visibility and safety for the staff and clients
 - b) To deter intruders and vandals
 - c) To record and store video footage for identification and prosecution
 - d) All of the above
5. Why is safe storage important in a veterinary clinic?
- a) To encourage theft and misuse of valuable items
 - b) To provide easy access to drugs, equipment, and records
 - c) To prevent loss or damage to sensitive items
 - d) To allow unrestricted access to unauthorized personnel

Part III: Give Short Answer Questions (6 points)

1. Define Principles of Veterinary Medical Ethics (2 points)?
2. List common security measures that can be incorporated in a veterinary clinic (2 Points)?
3. Briefly describe the best practices for stock control and inventory of veterinary drugs (2 Points)?

Notice: Scoring rates of above 8 is satisfactory and less than that is not satisfactory

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The experts who developed the learning guide

No	Name	Qualification	Educational background	Region	Phone number	E-mail
1	Dr. Dehinenet Terefe	DVM, MSc	Veterinary Medicine	Alage	0988043628	Dehinenetterefe2021@gmail.com
2	Dr. Sileshi Aregahagn	DVM, MSc	Veterinary Medicine	Kombolcha; Amhara	0920480599	kochasile@gmail.com
3	Dr. Yeshiwas Tarekegn	DVM, MSc	Veterinary Medicine	Alage	0910749374	yeshiwastarekegn@gmail.com
4	Dr. Boki Negasa	DVM, MSc	Veterinary Medicine	Holeta	0920563622	bokanegesa@gmail.com
5	Dr. Dejene Bayisa	DVM, MSc	Veterinary Medicine	Kombolcha Oromia	0921432254	dejebay@gmail.com
6	Dr. Bereket Addis	DVM, MSc	Veterinary Medicine	Alage	0912806054	bereketaddis2000@gmail.com